ASYMPTOTICS FOR RISK-PARAMETER ESTIMATION IN ZERO-DRIFT GJR-GARCH MODELS

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This paper develops statistical procedure to estimate risk parameter under a nonstationary zero-drift GJR-GARCH model. For a given risk measure r, the risk parameter is expressed as a function of the volatility coefficients and the risk of the innovation process. A one-step approach, replying on a reparameterization of the model, is considered and the use of a non-Gaussian QMLE is proposed. Consistency and asymptotic normality of the estimator of risk parameter are established for smooth risk measures as well as for the Value-at-Risk.

1. Introduction. Value-at-Risk (VaR) is a popular risk measure and plays an important role in modern quantitative risk management, see, e.g., Jorion (1996, 2006), Kuester, Mittnik and Paolella (2006), etc. The VaR is not coherent since it fails to fulfill the subadditivity property, see Artzner, et. al (1999). An alternative coherent risk measure related to the VaR is the Expected Shortfall (ES), also called the Expected Tail Loss, see Acerbi and Tasche (2002). Although they were commonly used last decades, the classical VaR and ES, which are based on the marginal distribution of returns, can not account for the dynamics of risk. Recently, in the literature there is an increasing interest in conditional risk measures (e.g., conditional VaR and conditional ES), which take into account the dynamics of risk and are widely used in risk management, see Gao and Song (2008), Yan, Shang and Linton (2013), Spierdijk (2016), Francq and Zakoïan (2018) and references therein. A comprehensive review on the VaR and the ES can be found in Jorion (2006) and Auer (2018).

To compare the performance of conditional risk estimator, Francq and Zakoïan (2015) introduced the concept of *risk parameter* in the framework of stationary conditional volatility models, which is a vector coefficient taking into account the dynamics of the returns and which can be interpreted as a summary of conditional risk. More specifically, consider a conditional volatility model of the form

$$\epsilon_t = \sigma_t(\vartheta_0)\eta_t,$$

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where $\{\epsilon_t\}$ is the log-return, $\sigma_t(\vartheta_0)$ a volatility process and ϑ_0 a finite-dimensional parameter, $\{\eta_t\}$ is a sequence of independent and identically distributed (i.i.d.) random variables, η_t is independent of $\{\epsilon_{t-j} : j \geq 1\}$. For a given risk measure r, which satisfies the positive homogeneity property (Artzner, et. al, 1999), such as the VaR or the ES, then the conditional risk of ϵ_t is given by

$$r_{t-1}(\epsilon_t) = \sigma_t(\vartheta_0) r(\eta_t),$$

where $r(\eta_t)$ is a constant and denotes the risk measure of the innovation. In most commonly used parametric GARCH-type models, multiplying the volatility by a constant amount to modifying the parameter value, see Table 1 in Francq and Zakoïan (2015). Under this assumption, then we have

$$r_{t-1}(\epsilon_t) = \sigma_t(\vartheta_0^*), \quad \text{where} \quad \vartheta_0^* = H(\vartheta_0, r(\eta_t))$$

for some specific function H. In this setting, ϑ_0^* is call the *risk parameter* associated with the risk function r. One of advantages of risk parameter is that it incorporates not only the volatility parameters but also the risk of the innovation process η_t.

Further, Francq and Zakoïan (2015) proposed two estimation procedures for risk parameter and compared their performance. One is the *two-step approach* based on the expression of ϑ_0^*. Under the identifiable condition $E\eta_t^2 = 1$, the Gaussian quasi-maximum likelihood estimator (QMLE) $\widehat{\vartheta}$ of ϑ_0 can be obtained, and then in turn an estimator $\hat{r}$ of $r(\eta_t)$ can be constructed from the residuals of the first step. Finally, the plug-in method yields an estimator $H(\widehat{\vartheta}, \hat{r})$ of the risk parameter ϑ_0^*.

The other, called the *one-step approach*, replies on a reparameterization of the conditional volatility model, defined as

$$\epsilon_t = \sigma_t(\vartheta_0^*)\eta_t^*, \quad \text{with} \quad r(\eta_t^*) = 1.$$

Here, the identifiable condition changes and replaces the standard assumption $E\eta_t^2 = 1$. The merit of such a representation is that the conditional risk $r_{t-1}(\epsilon_t)$ of ϵ_t can be estimated directly by $\sigma_t(\widehat{\vartheta}_0^*)$ if a consistent estimator $\widehat{\vartheta}_0^*$ of ϑ_0^* is obtained. Francq and Zakoïan (2015) demonstrated the advantages of the one-step approach.

In financial market, however, the log-returns of many financial assets may not be stationary. For this setting, nonstationary GARCH-type models are often used. Statistical inference for nonstationary GARCH-type models can be found in Nelson (1990), Jensen and Rahbek (2004a,b), Chan and Ng (2009), Linton, Pan and Wang (2010), Francq and Zakoïan (2012, 2013), Li, Li and Wu (2014), Li

and Wu (2018), etc. For estimation of nonstationary GARCH models, there is a dilemma that the intercept in volatility is not estimable so that they are not used to make prediction in practice. To tackle this dilemma, Li, et. al (2018) proposed a zero-drift GARCH model, in which the intercept is equal to zero, and applications demonstrate that a stable zero-drift GARCH(1, 1) model is more appropriate than a nonstationary GARCH(1, 1) model in fitting some nonstationary log returns. Similar studies can be found in Hafner and Preminger (2015) for a zero-drift ARCH(1) model and Li and Zhu (2018) for a zero-drift GJR-GARCH model.

The objective of the paper is to estimate the risk parameter in the zero-drift GJR-GARCH(1,1) model:

$$\begin{aligned} \epsilon_t &= \sigma_t \eta_t, \qquad r(\eta_t) = 1, \\ \sigma_t^2 &= \alpha_{0-}(\epsilon_{t-1}^-)^2 + \alpha_{0+}(\epsilon_{t-1}^+)^2 + \beta_0 \sigma_{t-1}^2, \end{aligned} \tag{1.1}$$

where $\{\eta_t\}$ is i.i.d. and r is a risk measure satisfying positive homogeneity (e.g., unconditional or conditional VaR and ES).

Model (1.1) is always nonstationary except for a trivial zero solution. Also, it can interpret the leverage effect in financial market through different values of α_{0-} and α_{0+}.

The paper is organized as follows. In Section 2 we discuss the QMLE of the risk parameter with its asymptotics. In Section 3 we present results of some numerical experiments, both of simulated and real data. Section 4 is devoted to the proofs of all results. Some technical proofs are collected in the Appendix.

2. Quasi-maximum likelihood estimation. Let $\theta_0 = (\alpha_{0-}, \alpha_{0+}, \beta)'$ and $\theta = (\alpha_-, \alpha_+, \beta)'$. Let $h(\cdot)$ be an arbitrary positive density which is called instrumental density, see Francq and Zakoïan (2015). Assume that the observed sample $\{\epsilon_1, \epsilon_2, ..., \epsilon_n\}$ is from model (1.1). Then the log-quasi-likelihood function is defined as

$$\bar{Q}_n(\theta) = \frac{1}{n} \sum_{t=1}^{n} l_t(\theta),$$

where

$$l_t(\theta) = g(\epsilon_t, \sigma_t(\theta)), \quad g(x, \sigma) = \log \frac{1}{\sigma} h(\frac{x}{\sigma})$$

and

$$\sigma_t^2(\theta) = \alpha_-(\epsilon_{t-1}^-)^2 + \alpha_+(\epsilon_{t-1}^+)^2 + \beta \sigma_{t-1}^2(\theta)$$

with the initial values ϵ_0 and $\sigma_t^2(\theta)$. Hereafter, we set $\epsilon_0 = \epsilon_0^*$ (a user-chosen non-zero constant) and $\sigma_t^2(\theta) = 0$ without loss of generality. In applications, we can always choose ϵ_0^* to be the first nonzero observation, and then do estimation for the remaining ones.

Our QMLE is

$$\hat{\theta}_n = \arg\max_{\theta\in\Theta} \bar{Q}_n(\theta), \tag{2.1}$$

where Θ is a compact subset of $\mathbb{R}^3$. When h is the standard Gaussian density, then the estimator is the standard Gaussian QMLE.

Note that in this setting, we can rewrite $l_t(\theta)$ as

$$l_t(\theta) = \log \frac{1}{\sigma_t(\theta)} h\left(\frac{\epsilon_t}{\sigma_t(\theta)}\right) = \log \frac{1}{\sigma_t(\theta)/\sigma_t} h\left(\frac{\eta_t}{\sigma_t(\theta)/\sigma_t}\right) - \log \sigma_t = g(\eta_t, \sigma_t(\theta)/\sigma_t) - \log \sigma_t$$

where σ_t is the realized value (with true parameter) in (1.1). For convenience of theoretical analysis, the maximization of $\tilde{Q}_n(\theta)$ is equivalent to that of

$$\widetilde{Q}_n(\theta) = \frac{1}{n}\sum_{t=1}^{n} g(\eta_t, \sigma_t(\theta)/\sigma_t) \tag{2.2}$$

where $\eta_t = \epsilon_t/\sigma_t$ are the realizations, and $\hat{\theta}_n = \arg\max_{\theta\in\Theta} \widetilde{Q}_n(\theta)$.

In practice, the choice of h heavily depends on that of risk measure r. An inappropriate choice of h will lead to inconsistent estimators of the risk parameter θ_0 and the conditional risk $r_{t-1}(\epsilon_t)$. Proposition 1 in Francq and Zakoïan (2015) summarizes the choice of h. Here we summarize the results in the following table.

Table 1
Examples of risk parameters and choices of h

Risk Parameter	$h(\cdot)$
$r(X) = (\mathbb{E}\lvert X\rvert^2)^{1/2}$	$h(x) \propto \exp(-\lvert x\rvert^2/2)$
$r(X) = (\mathbb{E}\lvert X\rvert^s)^{1/s}$, $1 \le s < 2$	$h(x) \propto \lvert x\rvert^{-(1-\lambda)} \exp(-\lambda\lvert x\rvert^s/s)$
$\mathrm{VaR}_\alpha(X)$	$h(x) = \lambda\alpha(1-2\alpha)\lvert x\rvert^{2\lambda\alpha-1}\{\lvert x\rvert^{-\lambda}\mathbf{1}_{\{\lvert x\rvert>1\}} + \mathbf{1}_{\{\lvert x\rvert\le 1\}}\}$

To establish asymptotic properties of $\hat{\theta}_n$, the following assumptions are needed.

Assumption 2.1. *$\mathbb{E}_{\theta_0} g(\eta_t, v)$ is uniquely maximized at $v = 1$.*

ASSUMPTION 2.2. *Θ is a compact subset of $(0,\infty)\times(0,\infty)\times[0,\infty)$.*

ASSUMPTION 2.3. *$g_1(x,v)=\frac{\partial}{\partial v}g(x,\sqrt{v})$ exists with $\mathbb{E}_{\theta_0}g_1^2(\eta_t,1)<+\infty$, i.e. the first order derivative of g is in L^2.*

ASSUMPTION 2.4. *$g_2(x,v)=\frac{\partial}{\partial v}g_1(x,v)$ exists and is continuous w.r.t. v, $\mathbb{E}|g_2(\eta_t,1)|<\infty$.*

ASSUMPTION 2.5. *h is continuous on $\mathbb{R}$, differentiable except on a finite set A, and there exist constants $\delta\geq 0$ and $C_0>0$ such that for all $u\in A^c$, $|uh'(u)/h(u)|\leq C_0(1+|u|^\delta)$ with $\mathbb{E}|\eta_t|^\delta<\infty$.*

THEOREM 2.1. *If Assumptions 2.1, 2.2, 2.5 hold, then $\hat{\theta}_n\to\theta_0$ a.s. as $n\to\infty$.*

THEOREM 2.2. *If Assumptions 2.1-2.5 hold, then*

$$\sqrt{n}(\hat{\theta}_n-\theta_0)\xrightarrow{d}N(0,\mathbb{E}g_1^2(\eta_t,1)A^{-1}\Sigma A^{-1}), \tag{2.3}$$

where the matrix A is given by

$$\begin{aligned}
A_{11}&=\mathbb{E}g_2(\eta_t,1)\mathbb{E}(v_t^{\alpha_-}(\theta_0))^2,\\
A_{12}=A_{21}&=\mathbb{E}g_2(\eta_t,1)\mathbb{E}[v_t^{\alpha_-}(\theta_0)v_t^{\alpha_+}(\theta_0)],\\
A_{13}=A_{31}&=\mathbb{E}g_2(\eta_t,1)\mathbb{E}[v_t^{\alpha_-}(\theta_0)v_t^{\beta}(\theta_0)]+\mathbb{E}g_1(\eta_t,1)\mathbb{E}v_t^{\beta\alpha_-}(\theta_0),\\
A_{22}&=\mathbb{E}g_2(\eta_t,1)\mathbb{E}(v_t^{\alpha_+}(\theta_0))^2,\\
A_{23}=A_{32}&=\mathbb{E}g_2(\eta_t,1)\mathbb{E}[v_t^{\alpha_+}(\theta_0)v_t^{\beta}(\theta_0)]+\mathbb{E}g_1(\eta_t,1)\mathbb{E}v_t^{\beta\alpha_+}(\theta_0),\\
A_{33}&=\mathbb{E}g_2(\eta_t,1)\mathbb{E}(v_t^{\beta}(\theta_0))^2+\mathbb{E}g_1(\eta_t,1)\mathbb{E}v_t^{\beta\beta}(\theta_0),
\end{aligned}$$

and

$$\Sigma=\begin{bmatrix}\mathbb{E}(v_t^{\alpha_-}(\theta_0))^2 & \mathbb{E}v_t^{\alpha_-}(\theta_0)v_t^{\alpha_+}(\theta_0) & \mathbb{E}v_t^{\alpha_-}(\theta_0)v_t^{\beta}(\theta_0)\\ \mathbb{E}v_t^{\alpha_-}(\theta_0)v_t^{\alpha_+}(\theta_0) & \mathbb{E}(v_t^{\alpha_+}(\theta_0))^2 & \mathbb{E}v_t^{\alpha_+}(\theta_0)v_t^{\beta}(\theta_0)\\ \mathbb{E}v_t^{\alpha_-}(\theta_0)v_t^{\beta}(\theta_0) & \mathbb{E}v_t^{\alpha_+}(\theta_0)v_t^{\beta}(\theta_0) & \mathbb{E}(v_t^{\beta}(\theta_0))^2\end{bmatrix},$$

with all values of expectations provided in Lemma A.1.

2.1. *Asymptotics for VaR estimator.* In the case of the VaR, the corresponding $h(\cdot)$ does not satisfy the continuity and differentiability assumptions. Therefore, Theorem 2.1 and 2.2 does not apply to VaR risk parameter. However, with another few assumptions, Assumptions 2.1 and 2.5 hold, yielding the asymptotic consistency as a corollary of Theorem 2.1. Also, with more assumptions the asymptotic normality of VaR holds.

For VAR $_\alpha(X) := -\sigma_t F_\eta^{-1}(\alpha)$, we define the instrumental density

$$h(x) = h_\alpha(x) = \lambda\alpha(1-2\alpha)|x|^{2\lambda\alpha-1}\{|x|^{-\lambda}\mathbf{1}_{\{|x|>1\}} + \mathbf{1}_{\{|x|\le 1\}}\} \tag{2.4}$$

with $\lambda = (2\alpha)^{-1}$ and $\hat{\theta}_n$ is given by Equation 2.1.

Assumption 2.6. *The distribution of η_t is nondegenerate on the event $\{|\eta_t| \ge 1\}$, i.e. $\{|\eta_t| \ge 1\} \ne \{|\eta_t| = \sigma\}$ for arbitrary $\sigma \ge 1$.*

Theorem 2.3. *If Assumptions 2.2 and 2.6 hold, and suppose the distribution of η_t is symmetric with $\mathbb{E}|\log|\eta_t|| < \infty$, then the QMLE $\hat{\theta}_n$ corresponding to Equations 2.2, 2.1 and 2.4 satisfies that $\hat{\theta}_n \to \theta_0$ a.s. as $n \to \infty$.*

Assumption 2.7. *The density function $f_\eta(x)$ of η_t is continuous at 1 and satisfies $f_\eta(1) > 0$ and $M = \sup_{x\in\mathbb{R}} |x| f_\eta(x) < \infty$.*

Then we have the following theorem

Theorem 2.4. *Under Assumption 2.7, for η_t with symmetric distribution, let $\hat{\theta}_n$ denote the estimator corresponding to the choice of h in Table 1 , then*

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, \frac{8\alpha(1-2\alpha)}{f^2_{\log|\eta_t|}(0)}\Sigma_\alpha^{-1}) \tag{2.5}$$

where

$$\Sigma_\alpha = \mathbb{E}Dv_t(\theta_0)Dv_t'(\theta_0) \tag{2.6}$$

at the true parameter θ_0 which is related to the risk level α. Here the notation $Dv_t(\theta) = \frac{\partial}{\partial\theta}v_t(\theta)$ is the vector of partial derivatives.

Theorem 2.4 also provides confidence interval for estimation of $\sigma_t(\theta)$. Denote $\Xi_\alpha := \frac{8\alpha(1-2\alpha)}{f^2_{\log|\eta_t|}(0)}\Sigma_\alpha^{-1}$. Suppose $\hat{\Xi}_\alpha$ is a consistent estimator of Ξ_α, then according to Delta method, the $(1-\alpha_0)$-confidence interval of $VaR_t(\theta_0)$, i.e. $\sigma_t(\theta_0)$ is bounded by

$$\sigma_t(\hat{\theta}_n) \pm \frac{1}{\sqrt{n}}\Phi^{-1}_{1-\alpha_0/2} \cdot \sqrt{\frac{\partial\sigma_t}{\partial\theta}(\hat{\theta}_n)' \cdot \hat{\Xi} \cdot \frac{\partial\sigma_t}{\partial\theta}(\hat{\theta}_n)} \tag{2.7}$$

where $\Phi^{-1}_{1-\alpha_0/2}$ is the $(1-\alpha_0/2)$ quantile of standard normal distribution.

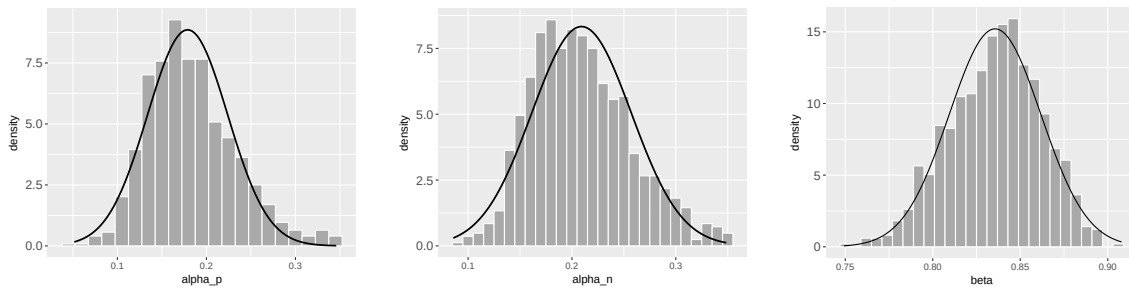
3. Numerical Experiments. In this section we carry out experiments both on simulated data and real data (nonstationary time series).

3.1. *Simulations.* Monte Carlo studies are carried out to see the performance of risk parameter estimators in finite sample. The simulations can be divided into the following two parts.

3.1.1. *Differentiable case.* For asymptotic consistency and normality of the differentiable case illustrated by Theorem 2.1 and Theorem 2.2, we consider risk parameter $r(X) = (\mathbb{E}|X|^2)^{1/2}$ and the corresponding estimator.

In our simulation, we first consider the underlying distribution of η_t being Gaussian such that $\mathbb{E}(\eta_t^2) = 1$. Specifically, we simulate with $\eta_t \sim N(0,1)$ and corresponding $g(x,v) = -\log v - \frac{x^2}{2v^2}$. It is easy to verify that the criterion g and the underlying distribution satisfy Assumptions 2.1, 2.3, 2.4 and 2.5. Therefore we expect the asymptotic consistency and normality of the estimator proposed in Equation 2.1. We generate K sets of time series data of length T, use the optimizer `nlminb` which requires only initial values in `R` to compute the estimator to throw light on our theoretical conclusions.

Empirical distribution. Fix $K = 1000$ and $T = 750$, with true values $\theta = (\alpha_+, \alpha_-, \beta)' = (0.150, 0.180, 0.855)'$ in which case σ_t oscillates between 0 and ∞, we obtain the following marginal empirical distributions of the estimators $\hat{\theta}$.



Distribution of $\hat{\alpha}_+$ Distribution of $\hat{\alpha}_-$ Distribution of $\hat{\beta}$

Fig 1: Distributions for $K = 1000$, $T = 750$, $\eta_t \sim N(0, 1)$.

The solid lines in the figures are densities of the Gaussian distribution whose mean value is the sample mean and variance is the variance indicated by Theorem 2.2. From the above figure it can be seen that the estimators $\hat{\theta}$ approximately follows a Gaussian distribution, while the mean value actually deviates a bit from θ.

Bias and MSEs. Now we look at the bias and mean squared errors (MSE) of our estimators. with different true values θ and different T but fixed $K = 1000$. For process $\eta_t \sim N(0, 1)$, we fix size of replicants $K = 1000$, let sample size $T = 300, 800, 1200$ and try three sets of true parameters corresponding to the nonstationary case.

Table 2
Bias and MSE (mean squared errors) of our estimator when $(\eta_t) \sim N(0, 1)$

Models		Parameters 1			Parameters 2			Parameters 3		
		α_+	α_-	β	α_+	α_-	β	α_+	α_-	β
		0.18	**0.24**	**0.821**	**0.5**	**0.6**	**0.607**	**0.7**	**0.6**	**0.555**
$T = 300$	Bias	0.006	0.005	-0.002	0.003	0.004	-0.006	0.006	0.009	0.002
	MSE	0.011	0.009	0.002	0.02	0.02	0.002	0.017	0.018	0.002
$T = 800$	Bias	0.004	0.003	-0.002	0.002	0.001	-0.005	0.001	0.003	-0.002
	MSE	0.006	0.005	0.001	0.005	0.005	0.001	0.009	0.007	0.001
$T = 1200$	Bias	0.003	0.003	-0.001	0.001	0.001	-0.001	0.003	0.004	-0.001
	MSE	0.004	0.004	0.001	0.001	0.001	0.001	0.005	0.005	0.000

The bias and MSEs of the estimators are summarized in Table 2. Our Monte Carlo experiments

reveal the proposed asymptotic consistency. The bias (mean errors) and MSEs (mean squared errors) decrease as sample size increases for all parameter settings. The finite sample behavior of the estimators agree with what we expect from asymptotic consistency and normality.

Coverage probability. According to the asymptotic distribution proposed in Theorem 2.2, we construct the approximate 95% confidence regions for the multivariate estimator $\hat{\theta}$ and see the coverage probability, i.e., the proportion of the K estimators falling in the confidence intervals.

We fix $K = 1000$, try different settings of true values $\theta = (\alpha_+, \alpha_-, \beta)'$ and sample sizes T. The asymptotic distribution of $\sqrt{n}(\hat{\theta}_n - \theta)$ is given in Equation 2.3. Specifially, denote $X = \sqrt{n}(\hat{\theta}_n - \theta)$, the $(1-\alpha)$ confidence region is defined by the set of X such that

$$X^T A \Sigma^{-1} A X \leq \mathbb{E} g_1^2(\eta_t, 1) \cdot \chi^2_{3,1-\alpha} = \frac{1}{2} \cdot \chi^2_{3,1-\alpha}$$

where $\chi^2_{3,1-\alpha}$ is the $(1-\alpha)$ quantile of χ^2_3.

Table 3
Coverage probabilities of our estimator when $(\eta_t) \sim N(0,1)$

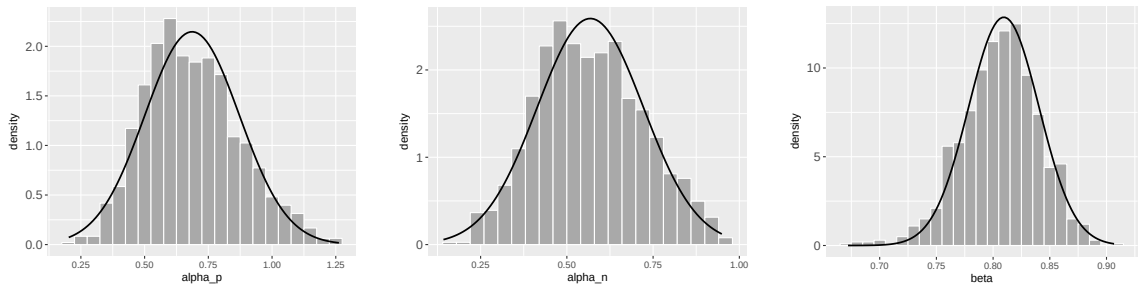
Models	Parameters 1			Parameters 2			Parameters 3		
	α_+	α_-	β	α_+	α_-	β	α_+	α_-	β
	0.18	**0.24**	**0.821**	**0.5**	**0.6**	**0.607**	**0.7**	**0.6**	**0.555**
$T = 300$	0.9316			0.9243			0.9134		
$T = 800$	0.9566			0.9336			0.9473		
$T = 1200$	0.9454			0.9493			0.9487		

The finite-sample results for $\alpha = 0.05$, i.e. the 95% coverage probability, are summarized in Table 5. We can see that when T is relatively small, a larger proportion of the estimators deviate from the true values. However when T is large, for example $T = 1200$, the 95% coverage probability is close to 0.95 in various parameter settings, indicating the asymptotic consistency as well as asymptotic normality we expect.

3.1.2. *VaR Case.* For the case of VaR illustrated by Theorem 2.3 and Theorem 2.4, we also repeatedly generate data and look at the distribution of the corresponding estimators.

Empirical distribution. Empirical distributions cast some light on our conclusions. Similarly, fix $K = 1000$, $T = 750$, for $\eta_t \sim \nu \cdot St_5$ where ν is such that $VaR_\alpha(\eta_t) = 1$, $\alpha = 0.05$, and true values

$\theta = (\alpha_+, \alpha_-, \beta)' = (0.7, 0.6, 0.8)$. K sets of data are generated and K estimators are computed. The following figures show the (marginal) empirical distributions, where the solid line is the density of Gaussian distribution with mean θ and variance indicated by Theorem 2.4.



Distribution of $\hat{\alpha}_+$ Distribution of $\hat{\alpha}_-$ Distribution of $\hat{\beta}$

Fig 2: Distributions for $K = 1000$, $T = 750$, $\eta_t \sim \nu \cdot St_5$.

The above empirical distributions shows the approximate normal distribution of the estimators.

Bias and MSEs. We also consider the biases and MSEs for different values of θ, T but fixed K.

TABLE 4
Bias and MSE (mean squared errors) of our estimator when $(\eta_t) \sim \nu \cdot St_5$

Models		Parameters 1			Parameters 2			Parameters 3		
		α_+	α_-	β	α_+	α_-	β	α_+	α_-	β
		0.18	**0.24**	**0.92**	**0.5**	**0.6**	**0.827**	**0.7**	**0.6**	**0.803**
$T = 300$	Bias	0.018	0.007	0.003	-0.016	-0.022	0.014	-0.017	0.007	0.011
	MSE	0.043	0.047	0.002	0.098	0.118	0.003	0.166	0.140	0.004
$T = 800$	Bias	-0.004	-0.002	0.002	0.006	-0.003	0.003	0.000	0.003	0.003
	MSE	0.007	0.009	0.001	0.031	0.036	0.001	0.047	0.043	0.003
$T = 1200$	Bias	-0.001	-0.001	0.001	-0.002	-0.004	0.002	0.004	0.005	0.001
	MSE	0.004	0.005	0.000	0.018	0.025	0.001	0.032	0.029	0.001

The above results are also in agreement with our asymptotic results. Both the bias and MSEs decrease as sample size T increases. However in some settings, the MSEs do not diminish to zero even when T is relatively large. The element $\hat{\beta}$ seems to converge better.

Coverage probability. Here we also consider the coverage probability for $\alpha = 0.05$. Still, we fix $K = 1000$ and try different settings of true values and sample sizes. According to the asymptotic distribution in Theorem 2.4, denote $X = \sqrt{n}(\hat{\theta}_n - \theta)$, the $(1-\alpha)$ confidence region is the set of X such that

$$X^T \Sigma_\alpha X \leq \frac{8\alpha(1-\alpha)}{f^2_{\log|\eta_t|}(0)} \cdot \chi^2_{3,1-\alpha}$$

with Σ_α defined in Theorem 2.4.

TABLE 5
Coverage probabilities of our estimator when $(\eta_t) \sim \nu \cdot St_5$

Models	Parameters 1			Parameters 2			Parameters 3		
	α_+	α_-	β	α_+	α_-	β	α_+	α_-	β
	0.18	**0.24**	**0.92**	**0.5**	**0.6**	**0.827**	**0.7**	**0.6**	**0.803**
$T = 300$	0.9360			0.9291			0.9333		
$T = 800$	0.9530			0.9490			0.9474		
$T = 1200$	0.9458			0.9498			0.9504		

The finite-sample results for $\alpha = 0.05$, i.e. the 95% coverage probability, are summarized in Table 5. We can see that when T is relatively small, the proportion of the estimators deviate largely from the true values is above α. However similar to the case $\eta_t \sim N(0,1)$, when T is large, the 95% coverage probability is close to 0.95 in various parameter settings. This agrees with our results for VaR case.

3.2. *Real data.* Expriments on real data are also conducted. We use stock prices data with non-stationary returns. Similar to experiments in Francq and Zakoïan (2012) and Li, et. al (2018), we study the daily stock data of Monarch Community Bancorp (NasdaqCM: MCBF), KV Pharmaceutical (NYSE: KV-A), Community Bankers Trust (AMEX: BTC) and China MediaExpress (NasdaqGS: CCME). According to the strict stationarity test, the log-return ($\times 100$) of each stock is non-stationary.

We fit the time series data of the stocks with our model (1.1) and plot the series σ_t. Meanwhile, we construct confidence intervals for σ_t's according to Equation (2.7) and plot the upper and lower bounds of the confidence intervals.

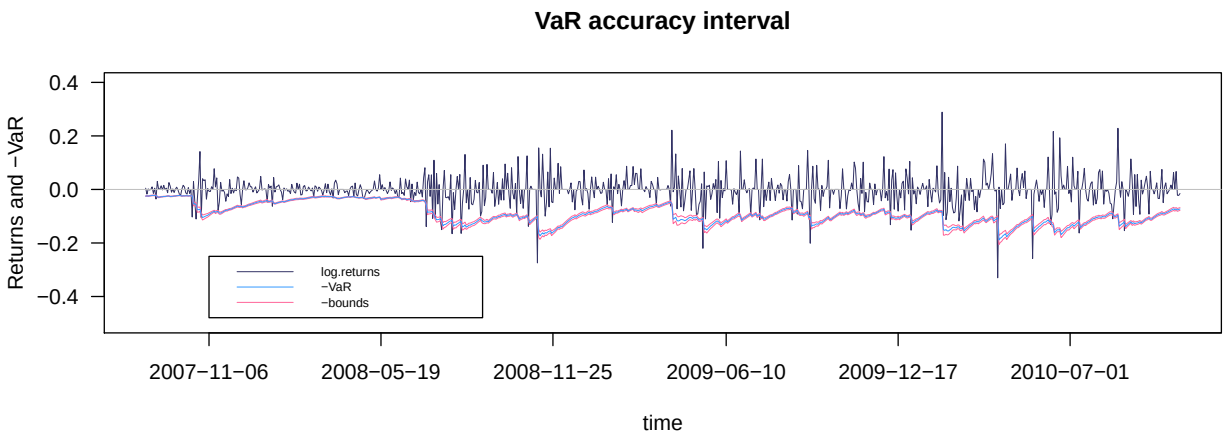


Fig 3: Log returns and Estimated -VaR$_t$ with $\alpha = 0.05$, as well as 0.95-confidence intervals of MCBF.

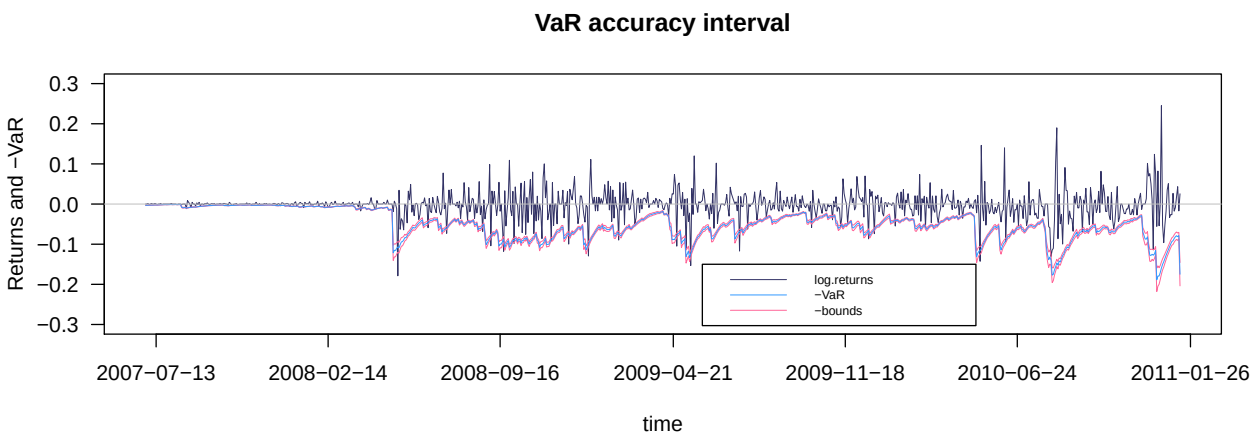


Fig 4: Log returns and Estimated -VaR$_t$ with $\alpha = 0.05$, as well as 0.95-confidence intervals of BTC.

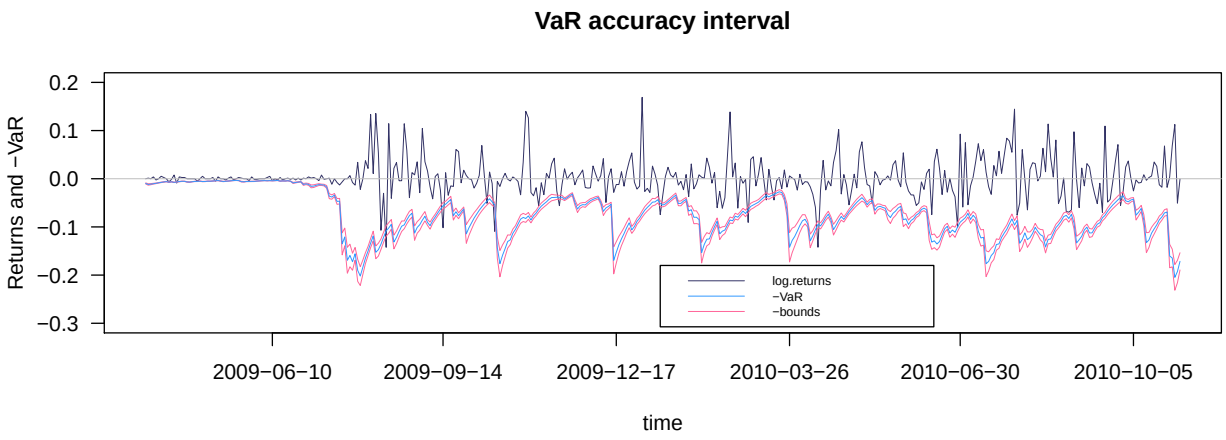


Fig 5: Log returns and Estimated -VaR$_t$ with $\alpha = 0.05$, as well as 0.95-confidence intervals of CCME.

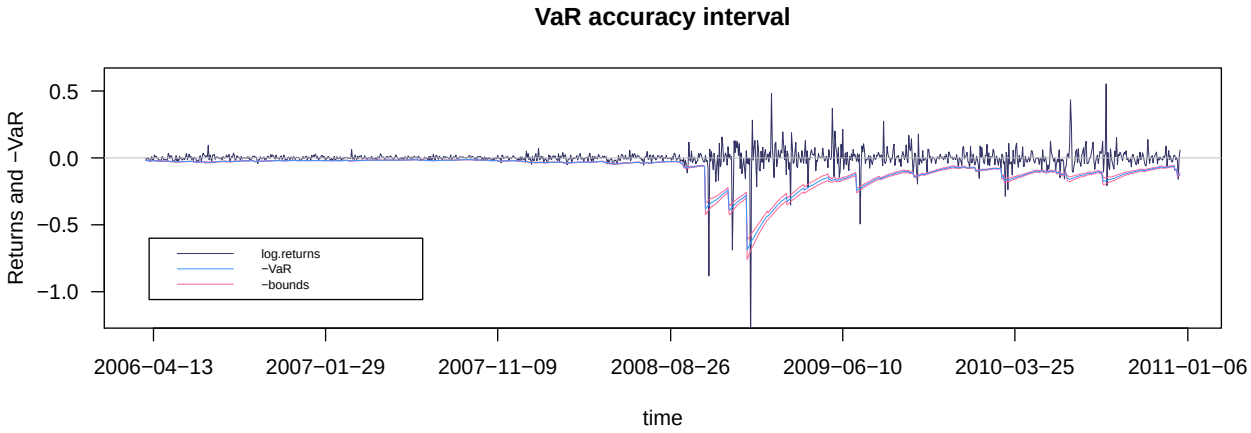


Fig 6: Log returns and Estimated -VaR$_t$ with $\alpha = 0.05$, as well as 0.95-confidence intervals of MCBF.

4. Proofs of Theorems.

4.1. *Approximation of Nonstationary Process $\sigma_t(\theta)$.* This plays a key role in proving the consistency and asymptotic normality of the estimator of the risk parameter.

PROPOSITION 4.1. *Suppose Assumption 2.2 holds. Let $a_0(x) = \beta_0 + \alpha_{0-}(x^-)^2 + \alpha_{0+}(x^+)^2$, and η_t, $t \in \mathbb{Z}_-$ are i.i.d. random variables with the same distribution as η_t, $t \geq 1$. For $\theta = (\alpha_-, \alpha_+, \beta)'$, define the process*

$$v_t(\theta) = \sum_{j=1}^{\infty} v_{tj}(\theta) = \sum_{j=1}^{\infty} \frac{\alpha_-(\eta_{t-j}^-)^2 + \alpha_+(\eta_{t-j}^+)^2}{a_0(\eta_{t-j})} \prod_{k=1}^{j-1} \frac{\beta}{a_0(\eta_{t-k})}. \tag{4.1}$$

Then $v_t(\alpha, \beta)$ is stationary and ergodic. Furthermore, let $\gamma_0 = \mathbb{E}\log[\beta_0 + \alpha_{0+}(\eta_t^+)^2 + \alpha_{0-}(\eta_t^-)^2]$, $\Theta_0 = \{\theta \in \Theta : \beta < e^{\gamma_0}\}$. Then for $\theta \in \Theta \backslash \Theta_0$, $\sigma_t^2(\theta)/\sigma_t^2 \to +\infty$ as $t \to \infty$. On the other hand, for each compact $\Theta^* \subset \Theta_0$, there exists a constant $c_0 > 0$ s.t.

(i)

$$\sup_{\theta \in \Theta^*} \left\{ e^{c_0 t} \left| \frac{\sigma_t^2(\theta)}{\sigma_t^2} - v_t(\theta) \right| \right\} \to 0 \quad a.s.$$

(ii)

$$\sup_{\theta \in \Theta^*} \left\{ e^{c_0 t} \left| \frac{\sigma_t^2}{\sigma_t^2(\theta)} - \frac{1}{v_t(\theta)} \right| \right\} \to 0 \quad a.s.$$

Moreover, $v_t(\alpha, \beta) = 1$ a.s. if and only if $\beta = \beta_0, \alpha_- = \alpha_{0-}, \alpha_+ = \alpha_{0+}$.

PROOF. Clearly, $v_t(\alpha, \beta)$ is stationary and ergodic. Note that

$$\frac{\sigma_t^2(\theta)}{\beta^t} = \frac{\alpha_+(\epsilon_{t-1}^+)^2 + \alpha_-(\epsilon_{t-1}^-)^2}{\beta^t} + \frac{\sigma_{t-1}^2(\theta)}{\beta^{t-1}},$$

we have

$$\sigma_t^2(\theta) = \sum_{k=1}^{t} \beta^{t-k}[\alpha_-(\epsilon_{t-k}^-)^2 + \alpha_+(\epsilon_{t-k}^+)^2]$$

and

$$\frac{\sigma_{t-k}^2}{\sigma_{t-k+1}^2} = \frac{\sigma_{t-k}^2}{\alpha_{0+}(\epsilon_{t-k}^+)^2 + \alpha_{0-}(\epsilon_{t-k}^-)^2 + \beta_0 \sigma_{t-k}^2} = \frac{1}{\beta_0 + \alpha_{0+}(\eta_{t-k}^+)^2 + \alpha_{0-}(\eta_{t-k}^-)^2} = \frac{1}{a_0(\eta_{t-k})}.$$

Thus,

$$\begin{aligned}\frac{\sigma_t^2(\theta)}{\sigma_t^2} &= \sum_{j=1}^{t} \beta^{j-1} \left\{ \prod_{k=1}^{j} \frac{\sigma_{t-k}^2}{\sigma_{t-k+1}^2} \right\} \frac{\alpha_-(\epsilon_{t-j}^-)^2 + \alpha_+(\epsilon_{t-j}^+)^2}{\sigma_{t-j}^2} \\ &= \sum_{j=1}^{t} \beta^{j-1} \left\{ \prod_{k=1}^{j} \frac{1}{a_0(\eta_{t-k})} \right\} \frac{\alpha_-(\epsilon_{t-j}^-)^2 + \alpha_+(\epsilon_{t-j}^+)^2}{\sigma_{t-j}^2} \\ &= \sum_{j=1}^{t} \frac{\alpha_+(\eta_{t-j}^+)^2 + \alpha_-(\eta_{t-j}^-)^2}{a_0(\eta_{t-j})} \prod_{k=1}^{j-1} \frac{\beta}{a_0(\eta_{t-k})}.\end{aligned}$$

It's easy to see that both $v_t(\theta)$ and $\sigma_t(\theta)/\sigma_t$ are positive since the distribution of η_t is non-degenerate, and increase with $\alpha_\pm$ and β. Using the expression of $v_t(\theta)$, we have

$$\begin{aligned}e^{ct}\left| v_t(\theta) - \frac{\sigma_t^2(\theta)}{\sigma_t^2} \right| &= \sum_{j=t+1}^{\infty} e^{ct} \frac{\alpha_-(\eta_{t-j}^-)^2 + \alpha_+(\eta_{t-j}^+)^2}{a_0(\eta_{t-j})} \prod_{k=1}^{j-1} \frac{\beta}{a_0(\eta_{t-k})} \\ &\le \sum_{j=t+1}^{\infty} \frac{\alpha_-(\eta_{t-j}^-)^2 + \alpha_+(\eta_{t-j}^+)^2}{a_0(\eta_{t-j})} \prod_{k=1}^{j-1} \frac{\beta \cdot e^c}{a_0(\eta_{t-k})} \\ &\le \sum_{j=t+1}^{\infty} \frac{\alpha_-(\eta_{t-j}^-)^2 + \alpha_+(\eta_{t-j}^+)^2}{a_0(\eta_{t-j})} \prod_{k=1}^{j-1} \frac{\bar{\beta} \cdot e^c}{a_0(\eta_{t-k})}\end{aligned}$$

where $\bar{\beta}$ is the supremum of β over the compact set Θ_0. Then letting $c = c_0 = \frac{1}{2}(\log \mathbb{E}(\beta_0 + \alpha_{0+}(\eta_t^+)^2 + \alpha_{0-}(\eta_t^-)^2) - \log \bar{\beta})$ we have $c > 0$. By the strong law of large numbers, we have

$$\frac{1}{j} \sum_{k=1}^{j-1} \log \frac{\bar{\beta} \cdot e^c}{a_0(\eta_{t-k})} \to \log \bar{\beta} + c - \log \mathbb{E}(\beta_0 + \alpha_{0+}(\eta_t^+)^2 + \alpha_{0-}(\eta_t^-)^2) < 0$$

Therefore by the Cauchy root test,

$$e^{c_0 t} \sup_{\theta \in \Theta^*} \left| \frac{\sigma_t^2(\theta)}{\sigma_t^2} - v_t(\theta) \right| \to 0 \quad a.s.$$

and $\sigma_t^2(\theta)/\sigma_t^2 \to \infty$ as $t \to \infty$ for $\theta \notin \Theta_0$.

On the other hand, since Θ is a compact subset of $(0,\infty) \times (0,\infty) \times [0,\infty)$, let $\underline{\alpha}_\pm = \inf\{\alpha_\pm \mid \theta \in \Theta\}$, $\overline{\alpha}_\pm = \sup\{\alpha_\pm \mid \theta \in \Theta\}$, $\underline{\beta} = \inf\{\beta \mid \theta \in \Theta\}$, $\overline{\beta} = \sup\{\beta \mid \theta \in \Theta\}$ we have $\underline{\alpha}_\pm > 0$. Letting $\underline{\theta} = (\underline{\alpha}_-, \underline{\alpha}_+, \underline{\beta})$ we have

$$e^{c_0 t} \left| \frac{\sigma_t^2}{\sigma_t^2(\theta)} - \frac{1}{v_t(\theta)} \right| = e^{c_0 t} \left| \frac{\sigma_t^2(\theta)}{\sigma_t^2} - v_t(\theta) \right| \cdot \left| \frac{\sigma_t^2}{v_t(\theta)\sigma_t^2(\theta)} \right| \le e^{c_0 t} \left| \frac{\sigma_t^2(\theta)}{\sigma_t^2} - v_t(\theta) \right| \cdot \left| \frac{\sigma_t^2}{v_t(\underline{\theta})\sigma_t^2(\underline{\theta})} \right|$$

Now since $\underline{\alpha}_{\pm} > 0$, we have $v_t(\underline{\theta}) > 0$ and $\sigma_t^2(\underline{\theta}) > 0$, thus

$$e^{c_0 t} \sup_{\theta \in \Theta^*} \left| \frac{\sigma_t^2}{\sigma_t^2(\theta)} - \frac{1}{v_t(\theta)} \right| \to 0 \quad a.s.$$

For the last statement, using the expression of $v_t(\alpha, \beta)$ again, simple calculation yields

$$\begin{aligned} v_{t-1}(\theta) &= \frac{1}{\beta} \sum_{j=2}^{\infty} \frac{\beta^{j-1}}{\prod_{k=2}^{j} a_0(\eta_{t-k})} [\alpha_-(\eta_{t-j}^-)^2 + \alpha_+(\eta_{t-j}^+)^2] \\ &= \frac{a_0(\eta_{t-1})}{\beta} \sum_{j=2}^{\infty} \frac{\beta^{j-1}}{\prod_{k=1}^{j} a_0(\eta_{t-k})} [\alpha_-(\eta_{t-j}^-)^2 + \alpha_+(\eta_{t-j}^+)^2] \\ &= \frac{a_0(\eta_{t-1})}{\beta} \left(v_t(\theta) - \frac{1}{a_0(\eta_{t-1})} [\alpha_-(\eta_{t-1}^-)^2 + \alpha_+(\eta_{t-1}^+)^2] \right) \\ &= \frac{a_0(\eta_{t-1})}{\beta} v_t(\theta) - \frac{1}{\beta} \left[\alpha_-(\eta_{t-1}^-)^2 + \alpha_+(\eta_{t-1}^+)^2 \right]. \end{aligned}$$

Then

$$(v_t(\theta) - 1) a_0(\eta_{t-1}) = \beta v_{t-1}(\theta) - \beta_0 + (\alpha_- - \alpha_{0-})(\eta_{t-1}^-)^2 + (\alpha_+ - \alpha_{0+})(\eta_{t-1}^+)^2$$

hence

$$v_t(\theta) = 1 \quad a.s. \quad \Longrightarrow \quad \beta = \beta_0, \alpha_- = \alpha_{0-}, \alpha_+ = \alpha_{0+}.$$

Also by the a.s. convergence of $\sigma_t^2(\theta)/\sigma_t^2$ to $v_t(\theta)$, we know $\theta = \theta_0$ implies $\sigma_t^2(\theta)/\sigma_t^2 = 1$ hence $v_t(\theta_0) = 1$ a.s., thus completing the proof. □

4.2. *Proof of Theorem 2.1.* Before the proof, we first give some lemmas.

Lemma 4.1. *Suppose Assumptions 2.2 and 2.5 hold. Define*

$$Q_n(\theta) = \frac{1}{n} \sum_{t=1}^{n} g(\eta_t, \sqrt{v_t(\theta)}) = \frac{1}{n} \sum_{t=1}^{n} \log \frac{1}{\sqrt{v_t(\theta)}} h\left(\frac{\eta_t}{\sqrt{v_t(\theta)}} \right) \tag{4.2}$$

then for each compact subset $\Theta^* \subset \Theta_0$, as $n \to \infty$ we have

$$\sup_{\theta \in \Theta^*} \left| Q_n(\theta) - \widetilde{Q}_n(\theta) \right| \to 0 \quad a.s.$$

Proof. Recall that

$$\begin{aligned}\left|Q_n(\theta)-\widetilde{Q}_n(\theta)\right| \leq & \frac{1}{n}\sum_{t=1}^{n}\left|\log\frac{1}{\sigma_t(\theta)/\sigma_t}h\left(\frac{\eta_t}{\sigma_t(\theta)/\sigma_t}\right)-\log\frac{1}{\sqrt{v_t(\theta)}}h\left(\frac{\eta_t}{\sqrt{v_t(\theta)}}\right)\right| \\ \leq & \frac{1}{2n}\sum_{t=1}^{n}\left|\log\frac{v_t(\theta)}{\sigma_t^2(\theta)/\sigma_t^2}\right|+\frac{1}{n}\sum_{t=1}^{n}\left|\log h\left(\frac{\eta_t}{\sigma_t(\theta)/\sigma_t}\right)-\log h\left(\frac{\eta_t}{\sqrt{v_t(\theta)}}\right)\right|\end{aligned}$$

Here the first term

$$\frac{1}{2n}\sum_{t=1}^{n}\left|\log\frac{v_t(\theta)}{\sigma_t^2(\theta)/\sigma_t^2}\right| \leq \frac{1}{2n}\sum_{t=1}^{n}\left|\frac{v_t(\theta)}{\sigma_t^2(\theta)/\sigma_t^2}-1\right| \leq \frac{1}{2n}\sum_{t=1}^{n}\frac{\sigma_t^2}{\sigma_t^2(\underline{\theta})}\left|v_t(\theta)-\frac{\sigma_t^2(\theta)}{\sigma_t^2}\right|$$

therefore by Proposition 4.1, it a.s. converges to zero uniformly over any compact subset $\Theta^* \subset \Theta_0$. For the second term, by Assumption 2.5 we have

$$\begin{aligned}&\frac{1}{n}\sum_{t=1}^{n}\left|\log h\left(\frac{\eta_t}{\sigma_t(\theta)/\sigma_t}\right)-\log h\left(\frac{\eta_t}{\sqrt{v_t(\theta)}}\right)\right| \\ =&\frac{1}{n}\sum_{t=1}^{n}\left|\frac{h'(\eta_t/u_t^*(\theta))}{h(\eta_t/u_t^*(\theta))}\right|\cdot\left|\frac{\sigma_t(\theta)}{\sigma_t}-\sqrt{v_t(\theta)}\right| \\ \leq&\frac{1}{n}\sum_{t=1}^{n}C_0\left|\frac{\eta_t}{\sigma_t(\underline{\theta})/\sigma_t}\right|^{\delta}\cdot\left|\frac{\sigma_t(\theta)}{\sigma_t}-\sqrt{v_t(\theta)}\right| \\ \leq&\frac{C_0}{n}\sum_{t=1}^{n}\left|\frac{\eta_t}{\sigma_t(\underline{\theta})/\sigma_t}\right|^{\delta}\cdot\frac{\sigma_t}{2\sigma_t(\underline{\theta})}\cdot\left|\frac{\sigma_t^2(\theta)}{\sigma_t^2}-v_t(\theta)\right| \\ =&\frac{C_0}{2n}\sum_{t=1}^{n}|\eta_t|^{\delta}\cdot\left|\frac{\sigma_t^2}{\sigma_t^2(\underline{\theta})}\right|^{\frac{1+\delta}{2}}\cdot\left|\frac{\sigma_t^2(\theta)}{\sigma_t^2}-v_t(\theta)\right|\end{aligned}$$

where $\sigma_t(\theta)/\sigma_t \leq u_t^*(\theta) \leq \sqrt{v_t(\theta)}$. Here

$$\frac{\sigma_t^2(\underline{\theta})}{\sigma_t^2} \geq \frac{\alpha_+(\eta_{t-1}^+)^2+\alpha_-(\eta_{t-1}^-)^2}{\beta_0+\alpha_{0+}(\eta_{t-1}^+)^2+\alpha_{0-}(\eta_{t-1}^-)^2} \geq \min\left\{\frac{\underline{\alpha}_+}{\alpha_{0+}},\frac{\underline{\alpha}_-}{\alpha_{0-}}\right\}>0$$

Since $\mathbb{E}|\eta_t|^{\delta}<\infty$, by Proposition 4.1 we know the term also a.s. converges to zero uniformly over $\theta \in \Theta^*$ for every compact $\Theta^* \subset \Theta_0$. Combining the preceding results we have that as $n \to \infty$,

$$\sup_{\theta\in\Theta^*}\left|\widetilde{Q}_n(\theta)-Q_n(\theta)\right| \to 0 \quad a.s.$$

□

LEMMA 4.2. *If Assumptions 2.1 and 2.2 hold, then for all* $\theta \neq \theta_0$, $\mathbb{E}_{\theta_0} g(\eta_t, v_t(\theta)) < \mathbb{E}_{\theta_0} g(\eta_t, 1) = \mathbb{E}_{\theta_0} g(\eta_t, \sqrt{v_t(\theta_0)})$.

PROOF. Since η_t is independent of $v_t(\theta)$, by conditional expectation it follows that

$$\mathbb{E}_{\theta_0} g(\eta_t, \sqrt{v_t(\theta)}) = \mathbb{E}_{\theta_0}\{\mathbb{E}_{\theta_0}[g(\eta_t, \sqrt{v_t(\theta)})|v_t(\theta)]\} \leq \mathbb{E}_{\theta_0}\{\mathbb{E}_{\theta_0}[g(\eta_t, 1)|v_t(\theta)]\} = \mathbb{E}_{\theta_0} g(\eta_t, 1)$$

The equality holds if and only if $v_t(\theta) = 1$ a.s. since $\mathbb{E}_{\theta_0} g(\eta_t, v)$ is uniquely maximized at $v = 1$. The latter equality is because $v_t(\theta_0) = 1$ a.s. □

LEMMA 4.3. *If Assumptions 2.1, 2.2 and 2.5 hold, then for any* $\theta \in \Theta_0 \backslash \{\theta_0\}$, *there exists a neighborhood* $V(\theta)$ *such that*

$$\limsup_{n\to\infty} \sup_{\theta^* \in V(\theta)} \widetilde{Q}_n(\theta^*) < \lim_{n\to\infty} \widetilde{Q}_n(\theta_0),$$

with $\widetilde{Q}_n(\theta)$ *defined in Equation 2.2.*

PROOF. Let $V_k(\theta)$ be the intersection of Θ with the ball centered at θ with radius $1/k$. Applying the ergodic theorem to $\left\{\sup_{\theta^* \in V_k(\theta)} g(\eta_t, \sqrt{v_t(\theta^*)})\right\}$ yields

$$\begin{aligned}
\limsup_{n\to\infty} \sup_{\theta^* \in V_k(\theta)} \widetilde{Q}_n(\theta^*) \leq & \limsup_{n\to\infty} \sup_{\theta^* \in V_k(\theta)} Q_n(\theta^*) + \limsup_{n\to\infty} \sup_{\theta^* \in V_k(\theta)} \left|\widetilde{Q}_n(\theta^*) - Q_n(\theta^*)\right| \\
= & \limsup_{n\to\infty} \sup_{\theta^* \in V_k(\theta^*)} \frac{1}{n} \sum_{k=1}^{n} g(\eta_t, \sqrt{v_t(\theta^*)}) \\
\leq & \limsup_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \sup_{\theta^* \in V_k(\theta)} g(\eta_t, \sqrt{v_t(\theta^*)}) \\
= & \mathbb{E}_{\theta_0} \left\{ \sup_{\theta^* \in V_k(\theta)} g(\eta_t, \sqrt{v_t(\theta^*)}) \right\}
\end{aligned}$$

Since $\left\{\sup_{\theta^* \in V_k(\theta)} g(\eta_t, \sqrt{v_t(\theta^*)})\right\}$ is bounded and decreases monotonely with k, by monotone convergence theorem, as $k \to \infty$,

$$\mathbb{E}_{\theta_0} \left\{ \sup_{\theta^* \in V_k(\theta)} g(\eta_t, \sqrt{v_t(\theta^*)}) \right\} \to \mathbb{E}_{\theta_0} g(\eta_t, \sqrt{v_t(\theta)})$$

Also, applying ergodic theorem to $g(\eta_t, \sqrt{v_t(\theta_0)}$ we have

$$\lim_{n\to\infty} \widetilde{Q}_n(\theta_0) \geq \lim_{n\to\infty} Q_n(\theta_0) - \limsup_{n\to\infty} \left|\widetilde{Q}_n(\theta_0) - Q_n(\theta_0)\right| = \mathbb{E}_{\theta_0} g(\eta_t, \sqrt{v_t(\theta_0)}) = \mathbb{E}_{\theta_0} g(\eta_t, 1)$$

Therefore there must exist some k and $V(\theta) := V_k(\theta)$ s.t.

$$\limsup_{n\to\infty} \sup_{\theta^* \in V(\theta)} \widetilde{Q}_n(\theta^*) < \lim_{n\to\infty} \widetilde{Q}_n(\theta_0).$$

□

Proof of Theorem 2.1. Note that $\widehat{\theta}_n = \arg\max_{\theta\in\Theta} \widetilde{Q}_n(\theta)$. Since for $\theta \notin \Theta_0$, $\sigma_t(\theta)/\sigma_t \to \infty$ a.s., it suffices to consider $\theta \in \Theta_0$. For any arbitrary neighborhood $V(\theta_0)$ of θ_0, it satisfies that

$$\liminf_{n\to\infty} \sup_{\theta^* \in V(\theta_0)} \widetilde{Q}_n(\theta^*) \geq \lim_{n\to\infty} \widetilde{Q}_n(\theta_0) = \mathbb{E}[g(\eta_t, 1)].$$

Also, by Lemma 4.3, for any $\theta \in V(\theta_0)^c$, there exists a $V(\theta)$ such that

$$\limsup_{n\to\infty} \sup_{\theta^* \in V(\theta)} \widetilde{Q}_n(\theta^*) < \lim_{n\to\infty} \widetilde{Q}_n(\theta_0).$$

Note that $V(\theta_0)$ and all $V(\theta)$'s form an open cover of compact set Θ. Then by compactness there exist $\theta_1, \ldots, \theta_k$ such that

$$\Theta \subset \cup_{i=0}^{k} V(\theta_i).$$

Then

$$\max_{\theta\in\Theta} \widetilde{Q}_n(\theta) = \max_{0\leq i\leq k} \max_{\theta\in V(\theta_i)} \widetilde{Q}_n(\theta).$$

Since for $i = 1, \ldots, k$,

$$\limsup_{n\to\infty} \sup_{\theta^* \in V_i(\theta)} \widetilde{Q}_n(\theta^*) < \lim_{n\to\infty} \widetilde{Q}_n(\theta_0),$$

for n large enough, $\hat{\theta}_n$ must lie in $V(\theta_0)$ (a.s.). By arbitrariness of $V(\theta_0)$, it follows that

$$\hat{\theta}_n \to \theta_0 \quad a.s.$$

The proof is complete. □

4.3. *Proof of Theorem 2.2.* Before the proof, we study the first- and second-order derivative processes of $v_t(\alpha,\beta)$ for asymptotic normality of $\widehat{\theta}_n$.

Note that

$$\frac{\sigma_t^2(\theta)}{\sigma_t^2}=\sum_{j=1}^{t}\beta^{j-1}\frac{\alpha_-(\eta_{t-j}^-)^2+\alpha_+(\eta_{t-j}^+)^2}{a_0(\eta_{t-j})}\prod_{k=1}^{j-1}\frac{1}{a_0(\eta_{t-k})},$$

by a direct calculation, we have that

$$\begin{aligned}
\frac{1}{\sigma_t^2}\frac{\partial\sigma_t^2(\theta)}{\partial\alpha_-}&=\sum_{j=1}^{t}\beta^{j-1}\frac{(\eta_{t-j}^-)^2}{a_0(\eta_{t-j})}\prod_{k=1}^{j-1}\frac{1}{a_0(\eta_{t-k})},\\
\frac{1}{\sigma_t^2}\frac{\partial\sigma_t^2(\theta)}{\partial\alpha_+}&=\sum_{j=1}^{t}\beta^{j-1}\frac{(\eta_{t-j}^+)^2}{a_0(\eta_{t-j})}\prod_{k=1}^{j-1}\frac{1}{a_0(\eta_{t-k})},\\
\frac{1}{\sigma_t^2}\frac{\partial\sigma_t^2(\theta)}{\partial\beta}&=\sum_{j=2}^{t}(j-1)\beta^{j-2}\frac{\alpha_-(\eta_{t-j}^-)^2+\alpha_+(\eta_{t-j}^+)^2}{a_0(\eta_{t-j})}\prod_{k=1}^{j-1}\frac{1}{a_0(\eta_{t-k})},\\
\frac{1}{\sigma_t^2}\frac{\partial^2\sigma_t^2(\theta)}{\partial\alpha_-^2}&=\frac{1}{\sigma_t^2}\frac{\partial^2\sigma_t^2(\theta)}{\partial\alpha_+^2}=\frac{1}{\sigma_t^2}\frac{\partial^2\sigma_t^2(\theta)}{\partial\alpha_-\partial\alpha_+}=\frac{1}{\sigma_t^2}\frac{\partial^2\sigma_t^2(\theta)}{\partial\alpha_+\partial\alpha_-}=0,\\
\frac{1}{\sigma_t^2}\frac{\partial^2\sigma_t^2(\theta)}{\partial\alpha_-\partial\beta}&=\sum_{j=2}^{t}(j-1)\beta^{j-2}\frac{(\eta_{t-j}^-)^2}{a_0(\eta_{t-j})}\prod_{k=1}^{j-1}\frac{1}{a_0(\eta_{t-k})},\\
\frac{1}{\sigma_t^2}\frac{\partial^2\sigma_t^2(\theta)}{\partial\alpha_+\partial\beta}&=\sum_{j=2}^{t}(j-1)\beta^{j-2}\frac{(\eta_{t-j}^+)^2}{a_0(\eta_{t-j})}\prod_{k=1}^{j-1}\frac{1}{a_0(\eta_{t-k})},\\
\frac{1}{\sigma_t^2}\frac{\partial^2\sigma_t^2(\theta)}{\partial\beta^2}&=\sum_{j=3}^{t}(j-1)(j-2)\beta^{j-3}\frac{\alpha_-(\eta_{t-j}^-)^2+\alpha_+(\eta_{t-j}^+)^2}{a_0(\eta_{t-j})}\prod_{k=1}^{j-1}\frac{1}{a_0(\eta_{t-k})}.
\end{aligned}$$

Let

$$\begin{aligned}
v_t(\theta)&=\sum_{j=1}^{\infty}\beta^{j-1}\frac{\alpha_-(\eta_{t-j}^-)^2+\alpha_+(\eta_{t-j}^+)^2}{a_0(\eta_{t-j})}\prod_{k=1}^{j-1}\frac{1}{a_0(\eta_{t-k})},\\
v_t^{\alpha_-}(\theta)&=\sum_{j=1}^{\infty}\beta^{j-1}\frac{(\eta_{t-j}^-)^2}{a_0(\eta_{t-j})}\prod_{k=1}^{j-1}\frac{1}{a_0(\eta_{t-k})},\\
v_t^{\alpha_+}(\theta)&=\sum_{j=1}^{\infty}\beta^{j-1}\frac{(\eta_{t-j}^+)^2}{a_0(\eta_{t-j})}\prod_{k=1}^{j-1}\frac{1}{a_0(\eta_{t-k})},\\
v_t^{\beta}(\theta)&=\sum_{j=1}^{\infty}(j-1)\beta^{j-2}\frac{\alpha_-(\eta_{t-j}^-)^2+\alpha_+(\eta_{t-j}^+)^2}{a_0(\eta_{t-j})}\prod_{k=1}^{j-1}\frac{1}{a_0(\eta_{t-k})},
\end{aligned}$$

$$v_t^{\beta\alpha_-}(\theta) = \sum_{j=2}^{\infty}(j-1)\beta^{j-2}\frac{(\eta_{t-j}^-)^2}{a_0(\eta_{t-j})}\prod_{k=1}^{j-1}\frac{1}{a_0(\eta_{t-k})},$$

$$v_t^{\beta\alpha_+}(\theta) = \sum_{j=2}^{\infty}(j-1)\beta^{j-2}\frac{(\eta_{t-j}^+)^2}{a_0(\eta_{t-j})}\prod_{k=1}^{j-1}\frac{1}{a_0(\eta_{t-k})},$$

$$v_t^{\beta\beta}(\theta) = \sum_{j=3}^{\infty}(j-1)(j-2)\beta^{j-3}\frac{\alpha_-(\eta_{t-j}^-)^2+\alpha_+(\eta_{t-j}^+)^2}{a_0(\eta_{t-j})}.\prod_{k=1}^{j-1}\frac{1}{a_0(\eta_{t-k})}$$

Similar to the argument in the proof of Proposition 4.1, it is not hard to prove the corresponding a.s. convergence as $n \to \infty$ (uniform over any compact set $\Theta^* \subset \Theta_0$).

For simplicity, let

$$\mu_1 = \mathbb{E}\frac{1}{a_0(\eta_t)}, \quad \mu_- = \mathbb{E}\frac{(\eta_t^-)^2}{a_0(\eta_t)}, \quad \mu_+ = \mathbb{E}\frac{(\eta_t^+)^2}{a_0(\eta_t)}.$$

Then,

$$\mathbb{E}v_t(\theta) = \frac{\alpha_+\mu_+ + \alpha_-\mu_-}{1-\beta\mu_1}$$

$$\mathbb{E}v_t^{\alpha_-}(\theta) = \frac{\mu_-}{1-\beta\mu_1}, \qquad \mathbb{E}v_t^{\alpha_+}(\theta) = \frac{\mu_+}{1-\beta\mu_1}, \qquad \mathbb{E}v_t^{\beta}(\theta) = \frac{\mu_1(\alpha_+\mu_+ + \alpha_-\mu_-)}{(1-\beta\mu_1)^2}$$

$$\mathbb{E}v_t^{\beta\alpha_-}(\theta) = \frac{\mu_1\mu_-}{(1-\beta\mu_1)^2}, \quad \mathbb{E}v_t^{\beta\alpha_+}(\theta) = \frac{\mu_1\mu_+}{(1-\beta\mu_1)^2}, \quad \mathbb{E}v_t^{\beta\beta}(\theta) = \frac{2\mu_1^2(\alpha_+\mu_+ + \alpha_-\mu_-)}{(1-\beta\mu_1)^3}.$$

Consider the first-order derivative of $\widetilde{Q}_n(\theta)$ at true θ_0, we have the following lemma

Lemma 4.4. *If Assumptions 2.1-2.4 hold, then*

$$\sqrt{n}\frac{\partial\widetilde{Q}_n(\theta_0)}{\partial\theta} \xrightarrow{d} N(0, \mathbb{E}g_1^2(\eta_t,1)\Sigma),$$

where

$$\Sigma = \begin{bmatrix} \mathbb{E}(v_t^{\alpha_-}(\theta_0))^2 & \mathbb{E}v_t^{\alpha_-}(\theta_0)v_t^{\alpha_+}(\theta_0) & \mathbb{E}v_t^{\alpha_-}(\theta_0)v_t^{\beta}(\theta_0) \\ \mathbb{E}v_t^{\alpha_-}(\theta_0)v_t^{\alpha_+}(\theta_0) & \mathbb{E}(v_t^{\alpha_+}(\theta_0))^2 & \mathbb{E}v_t^{\alpha_+}(\theta_0)v_t^{\beta}(\theta_0) \\ \mathbb{E}v_t^{\alpha_-}(\theta_0)v_t^{\beta}(\theta_0) & \mathbb{E}v_t^{\alpha_+}(\theta_0)v_t^{\beta}(\theta_0) & \mathbb{E}(v_t^{\beta}(\theta_0))^2 \end{bmatrix}.$$

PROOF. The first-order derivative of $\widetilde{Q}_n(\theta)$ at θ_0 is

$$\sqrt{n}\frac{\partial}{\partial\theta}\widetilde{Q}_n(\theta_0) = \frac{1}{\sqrt{n}}\sum_{t=1}^{n} g_1(\eta_t,1)\frac{\partial v_t(\theta_0)}{\partial\theta} + \frac{1}{\sqrt{n}}\sum_{t=1}^{n} g_1(\eta_t,1)\left[\frac{1}{\sigma_t^2}\frac{\partial\sigma_t^2(\theta)}{\partial\theta}\Big|_{\theta=\theta_0} - \frac{\partial v_t(\theta_0)}{\partial\theta}\right],$$

where $\partial v_t(\theta_0)/\partial\theta = (v_t^{\alpha_-}(\theta_0), v_t^{\alpha_+}(\theta_0), v_t^{\beta}(\theta_0))'$. Here

$$\frac{1}{\sqrt{n}}\sum_{t=1}^{n} g_1(\eta_t,1)\left[\frac{1}{\sigma_t^2}\frac{\partial\sigma_t^2(\theta)}{\partial\theta}\Big|_{\theta=\theta_0} - \frac{\partial}{\partial\theta}v_t(\theta_0)\right] = o_p(1).$$

Thus it suffices to show

$$\frac{1}{\sqrt{n}}\sum_{t=1}^{n} g_1(\eta_t,1)\frac{\partial v_t}{\partial\theta}(\theta_0) \xrightarrow{d} N(0, \mathbb{E}g_1^2(\eta_t,1)\Sigma)$$

By the Crámer-Wold device, it suffices to show that for any $\lambda = (\lambda_1,\lambda_2,\lambda_3)'$,

$$\lambda'\frac{1}{\sqrt{n}}\sum_{t=1}^{n} g_1(\eta_t,1)\frac{\partial v_t}{\partial\theta}(\theta_0) \xrightarrow{d} N(0, \mathbb{E}g_1^2(\eta_t,1)\lambda'\Sigma\lambda).$$

Note that

$$\lambda'\frac{1}{\sqrt{n}}\sum_{t=1}^{n} g_1(\eta_t,1)\frac{\partial v_t}{\partial\theta}(\theta_0) = \frac{1}{\sqrt{n}}\sum_{t=1}^{n} g_1(\eta_t,1)[\lambda_1 v_t^{\alpha_-}(\theta_0) + \lambda_2 v_t^{\alpha_+}(\theta_0) + \lambda_3 v_t^{\beta}(\theta_0)]$$

and $\{g_1(\eta_t,1)[\lambda_1 v_t^{\alpha_-}(\theta_0) + \lambda_2 v_t^{\alpha_+}(\theta_0) + \lambda_3 v_t^{\beta}(\theta_0)]\}$ is a martingale difference. Thus, by the Central Limit Theorem for martingale differences in Billingsley (1968)(p.476), we have

$$\frac{1}{\sqrt{n}}\sum_{t=1}^{n} g_1(\eta_t,1)[\lambda_1 v_t^{\alpha_-}(\theta_0) + \lambda_2 v_t^{\alpha_+}(\theta_0) + \lambda_3 v_t^{\beta}(\theta_0)] \xrightarrow{d} N(0, \mathbb{E}[g_1(\eta_t,1)]^2\lambda'\Sigma\lambda).$$

The proof is complete. □

PROOF OF THEOREM 2.2. By the Taylor expansion of $\frac{\partial}{\partial\theta}\widetilde{Q}_n(\theta)$ at $\hat{\theta}_n$, we have

$$0 = \sqrt{n}\frac{\partial\widetilde{Q}_n(\theta)}{\partial\theta} + \frac{\partial^2\widetilde{Q}_n(\theta^*)}{\partial\theta\partial\theta'}\sqrt{n}(\hat{\theta}_n - \theta_0),$$

where θ^* satisfies $\|\theta^* - \theta_0\| \le \|\widehat{\theta}_n - \theta_0\|$. Since $\widehat{\theta}_n \to \theta_0$ a.s. , we have $\theta^* \to \theta_0$ a.s. Denote

$$A_n(\theta) = \begin{bmatrix} \frac{\partial^2}{\partial\alpha_-^2}\widetilde{Q}_n(\theta) & \frac{\partial^2}{\partial\alpha_-\partial\alpha_+}\widetilde{Q}_n(\theta) & \frac{\partial^2}{\partial\alpha_-\partial\beta}\widetilde{Q}_n(\theta) \\ \frac{\partial^2}{\partial\alpha_-\partial\alpha_+}\widetilde{Q}_n(\theta) & \frac{\partial^2}{\partial\alpha_+^2}\widetilde{Q}_n(\theta) & \frac{\partial^2}{\partial\alpha_+\partial\beta}\widetilde{Q}_n(\theta) \\ \frac{\partial^2}{\partial\alpha_-\partial\beta}\widetilde{Q}_n(\theta) & \frac{\partial^2}{\partial\beta\partial\alpha_+}\widetilde{Q}_n(\theta) & \frac{\partial^2}{\partial\beta^2}\widetilde{Q}_n(\theta) \end{bmatrix}$$

By Assumption 2.4, it follows that

$$\frac{\partial^2 \widetilde{Q}_n(\theta)}{\partial\theta_i\partial\theta_j} = \frac{1}{n}\sum_{t=1}^{n} g_1\Big(\eta_t, \frac{\sigma_t^2(\theta)}{\sigma_t^2}\Big)\frac{1}{\sigma_t^2}\frac{\partial^2\sigma_t^2(\theta)}{\partial\theta_i\theta_j} + \frac{1}{n}\sum_{t=1}^{n} g_2\Big(\eta_t, \frac{\sigma_t^2(\theta)}{\sigma_t^2}\Big)\frac{1}{\sigma_t^2}\frac{\partial\sigma_t^2(\theta)}{\partial\theta_i}\frac{1}{\sigma_t^2}\frac{\partial\sigma_t^2(\theta)}{\partial\theta_j}.$$

By the law of large numbers for stationary and ergodic sequences, we have $A_n(\theta) \to A$, where

$$\begin{aligned}
A_{11} &= \mathbb{E}g_2(\eta_t, 1)(v_t^{\alpha_-}(\theta_0))^2,\\
A_{12} = A_{21} &= \mathbb{E}g_2(\eta_t, 1)v_t^{\alpha_-}(\theta_0)v_t^{\alpha_+}(\theta_0),\\
A_{13} = A_{31} &= \mathbb{E}g_2(\eta_t, 1)v_t^{\alpha_-}(\theta_0)v_t^{\beta}(\theta_0) + \mathbb{E}g_1(\eta_t, 1)v_t^{\beta\alpha_-}(\theta_0),\\
A_{22} &= \mathbb{E}g_2(\eta_t, 1)(v_t^{\alpha_+}(\theta_0))^2,\\
A_{23} = A_{32} &= \mathbb{E}g_2(\eta_t, 1)v_t^{\alpha_+}(\theta_0)v_t^{\beta}(\theta_0) + \mathbb{E}g_1(\eta_t, 1)v_t^{\beta\alpha_+}(\theta_0),\\
A_{33} &= \mathbb{E}g_2(\eta_t, 1)(v_t^{\beta}(\theta_0))^2 + \mathbb{E}g_1(\eta_t, 1)v_t^{\beta\beta}(\theta_0).
\end{aligned}$$

The values of A_{ij}'s are provided in Lemma A.1.

Thus,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N(0, \mathbb{E}g_1^2(\eta_t, 1)A^{-1}\Sigma A^{-1}).$$

The proof is complete. □

4.4. *Prrof of Theorem 2.3.* Just like the proof of Theorem 2.1, we first establish the following two lemmas.

Lemma 4.5. *Let $Q_n(\theta)$ be in Equation 4.2 with $h(\cdot)$ in Equation 2.4, and suppose Assumption 2.2 hold. Then for each compact set $\Theta^* \subset \Theta_0$, we have*

$$\sup_{\theta\in\Theta^*}\Big|Q_n(\theta) - \tilde{Q}_n(\theta)\Big| \to 0 \quad a.s.$$

Proof. By the form of $h(\cdot)$ and noting that $\sigma_t^2(\theta)/\sigma_t^2 \le v_t(\theta)$, we have

$$\begin{aligned}
&\Big|g(\eta_t, \frac{\sigma_t(\theta)}{\sigma_t}) - g(\eta_t, \sqrt{v_t(\theta)})\Big|\\
=&\Big|-\log\frac{\sigma_t(\theta)}{\sigma_t} + \log h(\frac{\eta_t}{\sigma_t(\theta)/\sigma_t}) + \log\sqrt{v_t(\theta)} - \log h(\frac{\eta_t}{\sqrt{v_t(\theta)}})\Big|
\end{aligned}$$

$$
\begin{aligned}
&\leq 2\lambda\alpha \left|\log\sqrt{v_t(\theta)} - \log\frac{\sigma_t(\theta)}{\sigma_t}\right| \\
&\quad + \lambda\left|\left(\log|\eta_t| - \log\frac{\sigma_t(\theta)}{\sigma_t}\right)\mathbf{1}_{[|\eta_t|>\sigma_t(\theta)/\sigma_t]} - \left(\log|\eta| - \log\sqrt{v_t(\theta)}\right)\mathbf{1}_{[|\eta_t|>\sqrt{v_t(\theta)}]}\right| \\
&= 2\lambda\alpha \left|\log\sqrt{v_t(\theta)} - \log\frac{\sigma_t(\theta)}{\sigma_t}\right| \\
&\quad + \lambda\left|\log\sqrt{v_t(\theta)} - \log\frac{\sigma_t(\theta)}{\sigma_t}\right| \cdot \mathbf{1}_{[|\eta_t|>\sqrt{v_t(\theta)}]} + \lambda\left|\log|\eta_t| - \log\frac{\sigma_t(\theta)}{\sigma_t}\right| \cdot \mathbf{1}_{[\sigma_t(\theta)/\sigma_t<|\eta_t|\leq\sqrt{v_t(\theta)}]} \\
&\leq \lambda(2\alpha+1)\left|\log\sqrt{v_t(\theta)} - \log\frac{\sigma_t(\theta)}{\sigma_t}\right|
\end{aligned}
$$

therefore by same arguments in the proof of Lemma 4.1 we obtain the desired result.

□

LEMMA 4.6. *Suppose Assumption 2.6 holds, and the distribution of η_t is symmetric with $\mathbb{E}|\log|\eta_t|| < \infty$. Let $g(x,\sigma)$ be defined with $h(\cdot)$ in Equation 2.4, then for each $\theta \neq \theta_0$, $\mathbb{E}_{\theta_0} g(\eta_t, v_t(\theta)) < \mathbb{E}_{\theta_0} g(\eta_t, 1) = \mathbb{E}_{\theta_0} g(\eta_t, v_t(\theta_0))$.*

PROOF. Since the distribution of η_t is symmetric and $r(\eta_t) = F_\eta^{-1}(1-\alpha) = 1$, we know $\mathbb{P}\{|\eta_t| > 1\} = 2\alpha$. Moreover,

$$
g(x,\sigma) - g(x,1) = -\lambda\left(\mathbf{1}_{[|x|>\sigma]} - \mathbf{1}_{[|x|>1]}\right)\log|x| - \lambda\left(2\alpha - \mathbf{1}_{[|x|>\delta]}\right)\log\sigma
$$

Therefore for $\sigma \neq 1$,

$$
\begin{aligned}
\mathbb{E}_{\theta_0} g(\eta_t,\sigma) - \mathbb{E}_{\theta_0} g(\eta_t,1) =& \mathbb{E}_{\theta_0}\left\{-\lambda\left(\mathbf{1}_{[|x|>\sigma]} - \mathbf{1}_{[|x|>1]}\right)\log|x| - \lambda\left(\mathbf{1}_{[|x|>1]} - \mathbf{1}_{[|x|>\delta]}\right)\log\sigma\right\} \\
=& \lambda\mathbb{E}_{\theta_0}\left\{\left(\mathbf{1}_{[|x|>1]} - \mathbf{1}_{[|x|>\sigma]}\right)\left(\log|x| - \log\sigma\right)\right\} \leq 1
\end{aligned}
$$

The inequality is strict for $\sigma \neq 1$ since the density for η_t is nondegenerate on the event $\{|\eta_t| \geq 1\}$. Therefore Assumption 2.1 holds, by same arguments in Lemma 4.2 we obtain the desired result. □

PROOF OF THEOREM 2.3. Since Lemmas 4.5 and 4.6 hold, following the same arguments in the proof of Lemma 4.3 and Theorem 2.1 we obtain the desired result.

□

PROOF OF THEOREM 2.4. Note that in the case of VaR, the corresponding criterion is

$$
g(\epsilon_t, \sigma_t(\theta)) = (2\lambda\alpha - 1)\log|\epsilon_t| - \lambda\mathbf{1}_{\{|\epsilon_t|>\sigma_t(\theta)\}}\log|\epsilon_t| - \lambda(2\alpha - \mathbf{1}_{\{|\epsilon_t|>\sigma_t(\theta)\}})\log\sigma_t(\theta) \tag{4.3}
$$

Under the assumptions of Theorem 2.4, the distribution of η_t is symmetric and if we denote $\rho_\alpha(u) = u(\alpha - \mathbf{1}_{\{u\le 0\}})$ proposed in Francq and Zakoïan (2015), we have

$$\hat{\theta}_n = \arg\min_{\theta\in\Theta} \frac{1}{n}\sum_{t=1}^{n} \rho_{1-2\alpha}\left(\log\frac{|\epsilon_t|}{\sigma_t(\theta)}\right) \tag{4.4}$$

Let $\hat{\nu}_n := \sqrt{n}(\hat{\theta}_n - \theta_0)$, then

$$\hat{\nu}_n = \arg\min_{\nu\in\sqrt{n}(\Theta-\theta_0)} \tilde{S}_n(\nu)$$

where

$$\tilde{S}_n(\nu) = \sum_{t=1}^{n} \rho_{1-2\alpha}\left\{\log\left(\frac{|\epsilon_t|}{\sigma_t(\theta_0 + n^{-1/2}\nu)}\right)\right\} - \sum_{t=1}^{n} \rho_{1-2\alpha}\left\{\log\left(\frac{|\epsilon_t|}{\sigma_t(\theta_0)}\right)\right\}$$

We will need several lemmas to complete the proof of Theorem 2.4.

Lemma 4.7. *Denote $Dv_t(\theta) = \frac{\partial}{\partial\theta} v_t(\theta) = (v_t^{\alpha-}(\theta), v_t^{\alpha+}(\theta), v_t^{\beta}(\theta))'$. Then for any bounded set Θ, we have*

$$\sup_{\nu\in\Theta} |\tilde{S}_n(\nu) - S_n(\nu)| = o_P(1)$$

where

$$S_n(\nu) = \sum_{t=1}^{n}\left(-\frac{\nu'}{2\sqrt{n}} Dv_t(\theta_0)(1 - 2\alpha - \mathbf{1}_{\{|\eta_t|<1\}}) + \int_0^{\frac{\nu'}{2\sqrt{n}} Dv_t(\theta_0)} \left(\mathbf{1}_{\{\log|\eta_t|\le s\}} - \mathbf{1}_{\{\log|\eta_t|<1\}}\right) ds\right)$$

proof. For simplicity, denote $D_t(\theta) = \frac{\partial}{\partial\theta}(\frac{\sigma_t^2(\theta)}{\sigma_t^2(\theta_0)})$, $Dv_t(\theta) = \frac{\partial}{\partial\theta} v_t(\theta)$, $\dot{D}_t(\theta) = \frac{\partial}{\partial\theta} D_t'(\theta)$. Clearly

$$\begin{aligned}
\tilde{S}_n(\nu) &= \sum_{t=1}^{n}\left(\rho_{1-2\alpha}\left\{\log\frac{|\epsilon_t|}{\sigma_t(\theta_0 + n^{-1/2}\nu)}\right\} - \rho_{1-2\alpha}\left\{\log\frac{|\epsilon_t|}{\sigma_t(\theta_0)}\right\}\right) \\
&= \sum_{t=1}^{n}\left(\rho_{1-2\alpha}\left\{\log\frac{|\eta_t|\sigma_t(\theta_0)}{\sigma_t(\theta_0 + n^{-1/2}\nu)}\right\} - \rho_{1-2\alpha}\left\{\log|\eta_t|\right\}\right) \\
&= \sum_{t=1}^{n}\left(\rho_{1-2\alpha}\left\{\log|\eta_t| - \frac{1}{2}\log\frac{\sigma_t^2(\theta_0 + \frac{\nu}{\sqrt{n}})}{\sigma_t^2(\theta_0)}\right\} - \rho_{1-2\alpha}\left\{\log|\eta_t|\right\}\right)
\end{aligned}$$

Use the identity

$$\rho_{1-2\alpha}(u - v) - \rho_{1-2\alpha}(u) = -v(1 - 2\alpha - \mathbf{1}_{\{u<0\}}) + \int_0^v \left(\mathbf{1}_{\{u\le s\}} - \mathbf{1}_{\{u<0\}}\right) ds$$

for $u \neq 0$ from Equation (A.3) in Koenker, Roger, et al. (2006). Therefore due to Taylor's expansion

$$\log \frac{\sigma_t^2(\theta_0 + \frac{\nu}{\sqrt{n}})}{\sigma_t^2(\theta_0)} = \frac{\nu'}{\sqrt{n}} D_t(\theta_0) + \frac{1}{2n} \nu' \dot{D}_t(\theta_t^*) \nu$$

where θ_t^* is on the line connecting θ_0 and $\theta_0 + \frac{\nu}{\sqrt{n}}$, we have

$$\begin{aligned}\tilde{S}_n(\nu) = &- \frac{1}{2} \sum_{t=1}^{n} \left(\frac{\nu'}{\sqrt{n}} D_t(\theta_0) + \frac{1}{2n} \nu' \dot{D}_t(\theta_t^*) \nu \right) \left(1 - 2\alpha - \mathbf{1}_{\{|\eta_t|<1\}}\right) \\ &+ \int_0^{\frac{1}{2}(\frac{\nu'}{\sqrt{n}} D_t(\theta_0) + \frac{1}{2n} \nu' \dot{D}_t(\theta_t^*) \nu)} \left(\mathbf{1}_{\{\log|\eta_t| \leq s\}} - \mathbf{1}_{\{\log|\eta_t|<0\}}\right) ds\end{aligned}$$

First we consider the term

$$\frac{1}{2n} \sum_{t=1}^{n} \nu' \dot{D}_t(\theta_t^*) \nu \left(1 - 2\alpha - \mathbf{1}_{\{|\eta_t|<1\}}\right)$$

Since θ_t^* is between θ_0 and $\theta_0 + \frac{\nu}{\sqrt{n}}$, we have $\theta_t^* \to \theta_0$ a.s. for any compact Θ. Also, due to the approximation to derivatives and the fact that $v_t^{\beta\alpha-}$, $v_t^{\beta\alpha+}$, $v_t^{\beta\beta}$ are stationary and ergodic, we have

$$\begin{aligned}&\sup_{\nu \in \Theta} \left| \frac{1}{2n} \sum_{t=1}^{n} \nu' \dot{D}_t(\theta_t^*) \nu \left(1 - 2\alpha - \mathbf{1}_{\{|\eta_t|<1\}}\right) \right| \\ \leq &\sup_{\nu \in \Theta} \left| \frac{1}{2n} \sum_{t=1}^{n} \nu' \frac{\partial^2 v}{\partial\theta\partial\theta'}(\theta_0) \nu \left(1 - 2\alpha - \mathbf{1}_{\{|\eta_t|<1\}}\right) \right| \\ &+ \sup_{\nu \in \Theta} \left| \frac{1}{2n} \sum_{t=1}^{n} \nu' \left(\frac{\partial^2 v}{\partial\theta\partial\theta'}(\theta_0) - \dot{D}_t(\theta_t^*) \right) \nu \left(1 - 2\alpha - \mathbf{1}_{\{|\eta_t|<1\}}\right) \right| = o_P(1).\end{aligned}$$

Now consider the term

$$\xi_{t,n}(\nu) := \int_{\frac{1}{2}\frac{\nu'}{\sqrt{n}} D_t(\theta_0)}^{\frac{1}{2}(\frac{\nu'}{\sqrt{n}} D_t(\theta_0) + \frac{1}{2n} \nu' \dot{D}_t(\theta_t^*) \nu)} \left(\mathbf{1}_{\{\log|\eta_t| \leq s\}} - \mathbf{1}_{\{\log|\eta_t|<0\}}\right) ds$$

We have the conditional expectation

$$\begin{aligned}|\mathbb{E}_{t-1} \xi_{t,n}(\nu)| &\leq \left| \int_{\frac{1}{2}\frac{\nu'}{\sqrt{n}} D_t(\theta_0)}^{\frac{1}{2}(\frac{\nu'}{\sqrt{n}} D_t(\theta_0) + \frac{1}{2n} \nu' \dot{D}_t(\theta_t^*) \nu)} \int_0^s f_{\log|\eta_t|}(u) du ds \right| \\ &\leq \left| \int_{\frac{1}{2}\frac{\nu'}{\sqrt{n}} D_t(\theta_0)}^{\frac{1}{2}(\frac{\nu'}{\sqrt{n}} D_t(\theta_0) + \frac{1}{2n} \nu' \dot{D}_t(\theta_t^*) \nu)} s \max_{u \in [0,s]} |f_{\log|\eta_t|}(u)| ds \right|\end{aligned}$$

$$
\begin{aligned}
&= \left| \int_{\frac{1}{2}\frac{\nu'}{\sqrt{n}} D_t(\theta_0)}^{\frac{1}{2}(\frac{\nu'}{\sqrt{n}} D_t(\theta_0) + \frac{1}{2n}\nu' \dot{D}_t(\theta_t^*)\nu)} s \max_{x\in[0,s]} \left| x f_{|\eta_t|}(x) \right| ds \right| \\
&\leq \frac{M}{8n} \nu' \dot{D}_t(\theta_t^*)\nu \left(\frac{\nu'}{\sqrt{n}} D_t(\theta_0) + \frac{1}{4n} \nu' \dot{D}_t(\theta_t^*)\nu \right)
\end{aligned}
$$

by Assumption 2.7. Therefore $|\sum_{t=1}^n \mathbb{E}_{t-1}\xi_{t,n}(\nu)| \leq O_P(n^{-1/2})$ uniformly over compact Θ and

$$
\sup_{\nu\in\Theta} \left| \sum_{t=1}^n \xi_{t,n}(\nu) \right| = o_P(1).
$$

So far we have proved that on compact set Θ,

$$
\sup_{\nu\in\Theta} |\tilde{S}_n(\nu) - \bar{S}_n(\nu)| = o_P(1)
$$

where

$$
\bar{S}_n(\nu) = \sum_{t=1}^n \left(-\frac{\nu'}{2\sqrt{n}} D_t(\theta_0)(1 - 2\alpha - \mathbf{1}_{\{|\eta_t|<1\}}) + \int_0^{\frac{\nu'}{2\sqrt{n}} D_t(\theta_0)} \left(\mathbf{1}_{\{\log|\eta_t|\leq s\}} - \mathbf{1}_{\{\log|\eta_t|<0\}} \right) ds \right)
$$

Moreover, by arguments in the approximation to derivatives, we know

$$
\sup_{\nu\in\Theta} \left\| \frac{1}{2} \sum_{t=1}^n \frac{\nu'}{\sqrt{n}} (D_t(\theta_0) - Dv_t(\theta_0)) \right\| = o_P(1)
$$

and

$$
\left| \mathbb{E}_{t-1} \int_{\frac{\nu'}{2\sqrt{n}} Dv_t(\theta_0)}^{\frac{\nu'}{2\sqrt{n}} D_t(\theta_0)} \left(\mathbf{1}_{\{\log|\eta_t|\leq s\}} - \mathbf{1}_{\{\log|\eta_t|<0\}} \right) ds \right| \leq \left| \int_{\frac{\nu'}{2\sqrt{n}} Dv_t(\theta_0)}^{\frac{\nu'}{2\sqrt{n}} D_t(\theta_0)} s \max_{u\in[0,s]} |f_{\log|\eta_t|}(u)| ds \right| = o_P(1)
$$

Therefore we have proved that on compact set Θ,

$$
\sup_{\nu\in\Theta} |S_n(\nu) - \bar{S}_n(\nu)| = o_P(1)
$$

where

$$
S_n(\nu) = \sum_{t=1}^n \left(-\frac{\nu'}{2\sqrt{n}} Dv_t(\theta_0)(1 - 2\alpha - \mathbf{1}_{\{|\eta_t|<1\}}) + \int_0^{\frac{\nu'}{2\sqrt{n}} Dv_t(\theta_0)} \left(\mathbf{1}_{\{\log|\eta_t|\leq s\}} - \mathbf{1}_{\{\log|\eta_t|<0\}} \right) ds \right)
$$

with $Dv_t(\theta)$ stationary and ergodic and belongs to the σ-field generated by $\{\eta_u \mid u < t\}$. Combine the two bounds and the proof is completed. □

LEMMA 4.8. *Under the same assumptions as Theorem 2.4, for any compact set Θ we have that $S_n(\cdot)$ converges weakly to $S(\cdot)$, where*

$$S(\nu) = \frac{1}{8} f_{\log|\eta_t|}(0)\nu'\Sigma_\alpha\nu + \frac{\sqrt{2\alpha(1-2\alpha)}}{2}\nu' N, \quad N \sim N(0, \Sigma_\alpha)$$

with $\Sigma_\alpha = \mathbb{E}Dv_t(\theta_0)Dv_t'(\theta_0)$, where θ_0 is related to risk level α.

Lemma 4.7 yields

$$S_n(\nu) = T_n(\nu) + U_n(\nu)$$

where

$$T_n(\nu) = -\frac{\nu'}{2\sqrt{n}} \sum_{t=1}^{n} Dv_t(\theta_0)(1 - 2\alpha - \mathbf{1}_{\{|\eta_t|<1\}})$$

and

$$U_n(\nu) = \sum_{t=1}^{n} \xi_{t,n}(\nu)$$

with

$$\xi_{t,n}(\nu) = \left(\log|\eta_t| - \frac{\nu'}{2\sqrt{n}} Dv_t(\theta_0)\right) \times \left\{\mathbf{1}_{\{\frac{\nu'}{2\sqrt{n}}Dv_t(\theta_0)<\log|\eta_t|<0\}} - \mathbf{1}_{\{\frac{\nu'}{2\sqrt{n}}Dv_t(\theta_0)>\log|\eta_t|>0\}}\right\}$$

We have the conditional expectation

$$\begin{aligned}
&\mathbb{E}_{t-1}\left(\log|\eta_t| - \frac{\nu'}{2\sqrt{n}} Dv_t(\theta_0)\right)\mathbf{1}_{\{\frac{\nu'}{2\sqrt{n}}Dv_t(\theta_0)>\log|\eta_t|>0\}} \\
=&\mathbf{1}_{\{\frac{\nu'}{2\sqrt{n}}Dv_t(\theta_0)>0\}} \int_0^{\frac{\nu'}{2\sqrt{n}}Dv_t(\theta_0)} \left(x - \frac{\nu'}{2\sqrt{n}} Dv_t(\theta_0)\right) f_{\log|\eta_t|}(x)dx \\
=&f_{\log|\eta_t|}(0)\mathbf{1}_{\{\frac{\nu'}{2\sqrt{n}}Dv_t(\theta_0)>0\}} \int_0^{\frac{\nu'}{2\sqrt{n}}Dv_t(\theta_0)} \left(x - \frac{\nu'}{2\sqrt{n}} Dv_t(\theta_0)\right) dx + R_{n,t} \\
=&-\frac{1}{8n} f_{\log|\eta_t|}(0)\mathbf{1}_{\{\frac{\nu'}{2\sqrt{n}}Dv_t(\theta_0)>0\}} \cdot \nu' Dv_t(\theta_0)Dv_t'(\theta_0)\nu + R_{n,t}
\end{aligned}$$

with

$$R_{n,t} = \mathbf{1}_{\{\frac{\nu'}{2\sqrt{n}}Dv_t(\theta_0)>0\}} \int_0^{\frac{\nu'}{2\sqrt{n}}Dv_t(\theta_0)} \left(x - \frac{\nu'}{2\sqrt{n}} Dv_t(\theta_0)\right)\left(f_{\log|\eta_t|}(x) - f_{\log|\eta_t|}(0)\right) dx$$

By continuity in Assumption 2.7, $\forall\epsilon > 0$, there exists $\delta > 0$ s.t. $|x| < \delta$ entails $|f_{\log|\eta_t|}(x) - f_{\log|\eta_t|}(0)| < \epsilon$. Therefore

$$|R_{n,t}| \le \frac{1}{8n}(\nu' Dv_t(\theta_0))^2(\epsilon + 2M \cdot \mathbf{1}_{\{\frac{\nu'}{2\sqrt{n}}Dv_t(\theta_0)>\delta\}})$$

By erogodity and stationarity of $Dv_t(\theta_0)$,

$$\sum_{t=1}^{n} \frac{1}{8n}(\nu' Dv_t(\theta_0))^2 \to \frac{1}{8}\mathbb{E}(\nu' Dv_t(\theta_0))^2 < \infty$$

as well as

$$\begin{aligned}
&\mathbb{E}\left|\sum_{t=1}^{n} \frac{1}{4n}(\nu' Dv_t(\theta_0))^2 \cdot \mathbf{1}_{\{\frac{\nu'}{2\sqrt{n}} Dv_t(\theta_0) > \delta\}}\right| \\
\leq &\frac{1}{4n}\sum_{t=1}^{n} \mathbb{E}\left|(\nu' Dv_t(\theta_0))^2 \cdot \mathbf{1}_{\{\nu' Dv_t(\theta_0) > 2\delta\sqrt{n}\}}\right| \\
\leq &\frac{1}{4n}\left(\sum_{t=1}^{N} \mathbb{E}\left|(\nu' Dv_t(\theta_0))^2\right| + \sum_{t=N+1}^{n} \mathbb{E}\left|(\nu' Dv_t(\theta_0))^2 \cdot \mathbf{1}_{\{\nu' Dv_t(\theta_0) > 2\delta\sqrt{N}\}}\right|\right) \\
\leq &o(1) + o(1)\mathbb{E}(\nu' Dv_t(\theta_0))^2 = o(1)
\end{aligned}$$

if we first take N s.t. $\mathbb{P}[\nu' Dv_t(\theta_0) > 2\delta\sqrt{N}]$ sufficiently small (to an arbitrarily small fixed value) then let $n \to \infty$. Therefore $\sum_{t=1}^{n} |R_{n,t}| = o_P(1)$. Similarly,

$$\begin{aligned}
&\mathbb{E}_{t-1}\left\{\left(\log|\eta_t| - \frac{\nu'}{2\sqrt{n}} Dv_t(\theta_0)\right)\mathbf{1}_{\{\frac{\nu'}{2\sqrt{n}} Dv_t(\theta_0) < \log|\eta_t| < 0\}}\right\} \\
=&\frac{1}{8n} f_{\log|\eta_t|}(0)\mathbf{1}_{\{\frac{\nu'}{2\sqrt{n}} Dv_t(\theta_0) < 0\}} \nu' Dv_t(\theta_0) Dv_t'(\theta_0)\nu + R_{n,t}'
\end{aligned}$$

with $\sum_{t=1}^{n} |R_{n,t}'| = o_P(1)$. Therefore we have

$$\begin{aligned}
\sum_{t=1}^{n} \mathbb{E}_{t-1}\xi_{t,n}(\nu) =& f_{\log|\eta_t|}(0)\frac{1}{8n}\sum_{t=1}^{n} \nu' Dv_t(\theta_0) Dv_t'(\theta_0)\nu + o_P(1) \\
=& f_{\log|\eta_t|}(0)\frac{1}{8}\nu'\mathbb{E}\{Dv_t(\theta_0) Dv_t'(\theta_0)\}\nu + o_P(1)
\end{aligned}$$

We employ the following elementary lemma

Lemma 4.9. *Suppose X is an integrable random variable and c is a real constant, then the random variable*

$$Y = (X - c)(\mathbf{1}_{\{c < X < 0\}} - \mathbf{1}_{\{c > X > 0\}})$$

satisfies $\mathrm{Var}Y \leq -c\mathbb{E}Y$.

to yield that the martingale difference $\bar{\xi}_{t,n}(\nu) := \xi_{t,n}(\nu) - \mathbb{E}_{t-1}\xi_{t,n}(\nu)$ satisfies $\sum_{t=1}^{n} \mathbb{E}_{t-1}\bar{\xi}_{t,n}^2(\nu) = o_P(1)$. Therefore

$$U_n(\nu) \xrightarrow{P} f_{\log|\eta_t|}(0)\frac{1}{8}\nu'\mathbb{E}\{Dv_t(\theta_0)Dv_t'(\theta_0)\}\nu$$

Also, the CLT for martingale yields that

$$T_n(\nu) \xrightarrow{d} \frac{\sqrt{2\alpha(1-2\alpha)}}{2}\nu' N, \quad N \sim N(0, \mathbb{E}Dv_t(\theta_0)Dv_t'(\theta_0)) = N(0, \Sigma_\alpha)$$

Note that θ_0 here is related to the risk level α.

So far it has been proved that finite dimensional joint distributions of $S_n(\cdot)$ converge to that of $S(\cdot)$. Since the path of $S_n(\cdot)$ is convex, we have S_n converges weakly to the process $S(\cdot)$.

□

Now we return to complete the proof of Theorem 2.4.

PROOF OF THEOREM 2.4. In Lemma 4.7 and Lemma 4.8, we know $\tilde{S}_n(\nu)$ converges weakly to the process

$$S(\nu) = \frac{1}{8}f_{\log|\eta_t|}(0)\nu'\Sigma_\alpha\nu + \frac{\sqrt{2\alpha(1-2\alpha)}}{2}\nu' N, \quad N \sim N(0, \Sigma_\alpha)$$

which is minimized at

$$\hat{\nu} = -\frac{2\sqrt{2\alpha(1-2\alpha)}}{f_{\log|\eta_t|}(0)}\Sigma_\alpha^{-1}N \sim N(0, \frac{8\alpha(1-2\alpha)}{f^2_{\log|\eta_t|}(0)}\Sigma_\alpha^{-1})$$

Since $S_n(\nu)$ is convex for all ν, Remark 1 and Lemma 2.2 in Davis, et. al (1992) yields the convergence of the minimizer $\sqrt{n}(\hat{\theta} - \theta_0)$. □

APPENDIX A: SOME LEMMAS

To state results conveniently, we first list the first- and second-derivatives of process $v_t(\alpha, \beta)$:

$$v_t(\theta) = \sum_{j=1}^{\infty} \beta^{j-1}\frac{\alpha_-(\eta_{t-j}^-)^2 + \alpha_+(\eta_{t-j}^+)^2}{a_0(\eta_{t-j})}\prod_{i=1}^{j-1}\frac{1}{a_0(\eta_{t-k})},$$

$$v_t^{\alpha-}(\theta) = \sum_{j=1}^{\infty} \beta^{j-1}\frac{(\eta_{t-j}^-)^2}{a_0(\eta_{t-j})}\prod_{i=1}^{j-1}\frac{1}{a_0(\eta_{t-k})},$$

$$
\begin{aligned}
v_t^{\alpha+}(\theta) &= \sum_{j=1}^{\infty} \beta^{j-1} \frac{(\eta_{t-j}^{+})^2}{a_0(\eta_{t-j})} \prod_{i=1}^{j-1} \frac{1}{a_0(\eta_{t-k})}, \\
v_t^{\beta}(\theta) &= \sum_{j=1}^{\infty} (j-1)\beta^{j-2} \frac{\alpha_-(\eta_{t-j}^{-})^2 + \alpha_+(\eta_{t-j}^{+})^2}{a_0(\eta_{t-j})} \prod_{i=1}^{j-1} \frac{1}{a_0(\eta_{t-k})}, \\
v_t^{\beta\alpha-}(\theta) &= \sum_{j=2}^{\infty} (j-1)\beta^{j-2} \frac{(\eta_{t-j}^{-})^2}{a_0(\eta_{t-j})} \prod_{i=1}^{j-1} \frac{1}{a_0(\eta_{t-k})}, \\
v_t^{\beta\alpha+}(\theta) &= \sum_{j=2}^{\infty} (j-1)\beta^{j-2} \frac{(\eta_{t-j}^{+})^2}{a_0(\eta_{t-j})} \prod_{i=1}^{j-1} \frac{1}{a_0(\eta_{t-k})}, \\
v_t^{\beta\beta}(\theta) &= \sum_{j=3}^{\infty} (j-1)(j-2)\beta^{j-3} \frac{\alpha_-(\eta_{t-j}^{-})^2 + \alpha_+(\eta_{t-j}^{+})^2}{a_0(\eta_{t-j})} \prod_{i=1}^{j-1} \frac{1}{a_0(\eta_{t-k})}.
\end{aligned}
$$

Further, we define

$$
\begin{aligned}
\nu_1 &= \mathbb{E}\frac{1}{a_0^2(\eta_t)}, \quad \nu_- = \mathbb{E}\frac{(\eta_t^-)^2}{a_0^2(\eta_t)}, \quad \nu_+ = \mathbb{E}\frac{(\eta_t^+)^2}{a_0^2(\eta_t)}, \\
\delta_- &= \mathbb{E}\frac{(\eta_t^-)^4}{a_0^2(\eta_t)}, \quad \delta_+ = \mathbb{E}\frac{(\eta_t^+)^4}{a_0^2(\eta_t)}, \quad \delta_{+-} = \mathbb{E}\frac{(\eta_t^+)^2(\eta_t^-)^2}{a_0^2(\eta_t)},
\end{aligned}
$$

and

$$
\mu_1 = \mathbb{E}\frac{1}{a_0(\eta_t)}, \quad \mu_- = \mathbb{E}\frac{(\eta_t^-)^2}{a_0(\eta_t)}, \quad \mu_+ = \mathbb{E}\frac{(\eta_t^+)^2}{a_0(\eta_t)}.
$$

Lemma A.1. *With aforementioned notation, it follows that*

$$
\begin{aligned}
\mathbb{E}(v_t^{\alpha-}(\theta))^2 &= \frac{\delta_-}{1-\beta^2\nu_1} + \frac{2\beta\mu_-\nu_-}{(1-\beta\mu_1)(1-\beta\nu_1)}, \\
\mathbb{E}v_t^{\alpha-}(\theta_0)v_t^{\alpha+}(\theta) &= \frac{\delta_{+-}}{1-\beta^2\nu_1} + \frac{\beta\mu_-\nu_-}{(1-\beta\mu_1)(1-\beta^2\nu_1)} + \frac{\beta\nu_-\mu_+\mu_1^2}{(1-\beta\mu_1)(1-\beta^2\nu_1)}, \\
\mathbb{E}v_t^{\alpha-}(\theta_0)v_t^{\beta}(\theta) &= \frac{\nu_1\beta(\alpha_+\delta_{+-} + \alpha_-\delta_-)}{(1-\beta^2\nu_1)^2} + \frac{(\alpha_+\nu_+ + \alpha_-\nu_-)\mu_-\beta^2\nu_1}{(1-\beta\mu_1)(1-\beta^2\nu_1)^2}, \\
&\quad + \frac{\mu_1^3\nu_-(\alpha_+\mu_+ + \alpha_-\mu_-)}{(\mu_1-\beta\nu_1)(1-\beta\mu_1)^2} - \frac{\mu_1^2\nu_-\beta\nu_1(\alpha_+\mu_+ + \alpha_-\mu_-)}{(1-\beta\mu_1)(1-\beta^2\nu_1)^2} \\
\mathbb{E}(v_t^{\alpha+}(\theta))^2 &= \frac{\delta_+}{1-\beta^2\nu_1} + \frac{2\beta\mu_+\nu_+}{(1-\beta\mu_1)(1-\beta\nu_1)}, \\
\mathbb{E}v_t^{\alpha+}(\theta_0)v_t^{\beta}(\theta) &= \frac{\nu_1\beta(\alpha_+\delta_+ + \alpha_-\delta_{+-})}{(1-\beta^2\nu_1)^2} + \frac{(\alpha_+\nu_+ + \alpha_-\nu_-)\mu_+\beta^2\nu_1}{(1-\beta\mu_1)(1-\beta^2\nu_1)^2},
\end{aligned}
$$

$$+\frac{\mu_1^3\nu_+(\alpha_+\mu_+ + \alpha_-\mu_-)}{(\mu_1-\beta\nu_1)(1-\beta\mu_1)^2} - \frac{\mu_1^2\nu_+\beta\nu_1(\alpha_+\mu_+ + \alpha_-\mu_-)}{(1-\beta\mu_1)(1-\beta^2\nu_1)^2},$$

$$\mathbb{E}(v_t^\beta(\theta))^2 = (\alpha_-^2\delta_- + \alpha_+^2\delta_+ + 2\alpha_+\alpha_-\delta_{+-})\frac{\nu_1(1+\beta^2\nu_1)}{(1-\beta^2\nu_1)^3},$$

$$+2(\alpha_-\nu_- + \alpha_+\nu_+)(\alpha_-\mu_- + \alpha_+\mu_+)\Big[\frac{\beta^2\mu_1\nu_1}{(1-\beta^2\nu_1)^2(1-\mu_1\beta)^2} + \frac{2\beta^2\nu_1}{(1-\mu_1\beta)(1-\beta^2\nu_1)^3}\Big].$$

Proof. By a direct calculation, we have that

$$\begin{aligned}
\mathbb{E}(v_t^{\alpha_-}(\theta))^2 =& \sum_{i=1}^\infty\sum_{j=1}^\infty \mathbb{E}\beta^{j-1}\frac{(\eta_{t-j}^-)^2}{a_0(\eta_{t-j})}\frac{1}{a_0(\eta_{t-1})\cdots a_0(\eta_{t-j+1})}\beta^{i-1}\frac{(\eta_{t-i}^-)^2}{a_0(\eta_{t-i})}\frac{1}{a_0(\eta_{t-1})\cdots a_0(\eta_{t-i+1})}\\
=& \sum_{j=1}^\infty \mathbb{E}\beta^{2j-2}\frac{(\eta_{t-j}^-)^4}{a_0(\eta_{t-j})^2}\frac{1}{a_0(\eta_{t-1})^2\cdots a_0(\eta_{t-j+1})^2}\\
&+\sum_{i=1}^\infty\sum_{j>i}\beta^{i+j-2}\mathbb{E}\frac{(\eta_{t-j}^-)^2}{a_0(\eta_{t-j})}\frac{(\eta_{t-i}^-)^2}{a_0(\eta_{t-i})^2}\frac{1}{a_0(\eta_{t-1})^2\cdots a_0(\eta_{t-i+1})^2}\cdot\frac{1}{a_0(\eta_{t-i-1})\cdots a_0(\eta_{t-j+1})}\\
=& \sum_{j=1}^\infty \delta_-(\beta^2\nu_1)^{j-1} + 2\sum_{i=1}^\infty\sum_{j>i}\beta^{j+i-2}\mu_-\nu_-\nu_1^{i-1}\mu_1^{j-i-1}\\
=& \frac{\delta_-}{1-\beta^2\nu_1} + \frac{2\beta\mu_-\nu_-}{(1-\beta\mu_1)(1-\beta^2\nu_1)}.
\end{aligned}$$

Secondly,

$$\begin{aligned}
\mathbb{E}v_t^{\alpha_-}(\theta_0)v_t^{\alpha_+}(\theta) =& \sum_{i=1}^\infty\sum_{j=1}^\infty \mathbb{E}\beta^{j-1}\frac{(\eta_{t-j}^-)^2}{a_0(\eta_{t-j})}\frac{1}{a_0(\eta_{t-1})\cdots a_0(\eta_{t-j+1})}\beta^{i-1}\frac{(\eta_{t-i}^+)^2}{a_0(\eta_{t-i})}\frac{1}{a_0(\eta_{t-1})\cdots a_0(\eta_{t-i+1})}\\
=& \sum_{j=1}^\infty \mathbb{E}\beta^{2j-2}\frac{(\eta_{t-j}^-)^2(\eta_{t-j}^+)^2}{a_0(\eta_{t-j})^2}\frac{1}{a_0(\eta_{t-1})^2\cdots a_0(\eta_{t-j+1})^2}\\
&+\sum_{i=1}^\infty\sum_{j>i}\beta^{i+j-2}\mathbb{E}\frac{(\eta_{t-j}^-)^2}{a_0(\eta_{t-j})}\frac{(\eta_{t-i}^+)^2}{a_0(\eta_{t-i})^2}\frac{1}{a_0(\eta_{t-1})^2\cdots a_0(\eta_{t-i+1})^2}\cdot\frac{1}{a_0(\eta_{t-i-1})\cdots a_0(\eta_{t-j+1})}\\
&+\sum_{i=1}^\infty\sum_{j=1}^{i-1}\beta^{i+j-2}\mathbb{E}\frac{(\eta_{t-j}^-)^2}{a_0(\eta_{t-j})^2}\frac{(\eta_{t-i}^+)^2}{a_0(\eta_{t-i})}\frac{1}{a_0(\eta_{t-1})^2\cdots a_0(\eta_{t-j+1})^2}\cdot\frac{1}{a_0(\eta_{t-j-1})\cdots a_0(\eta_{t-i+1})}\\
=& \frac{\delta_{+-}}{1-\beta^2\nu_1} + \sum_{i=1}^\infty\sum_{j>i}\beta^{i+j-2}\mu_-\nu_+\nu_1^{i-1}\mu_1^{j-i-1} + \sum_{i=2}^\infty\sum_{j=1}^{i-1}\beta^{i+j-2}\nu_-\mu_+\nu_1^{j-1}\mu_1^{i-j-1}\\
=& \frac{\delta_{+-}}{1-\beta^2\nu_1} + \frac{\beta\mu_-\nu_-}{(1-\beta\mu_1)(1-\beta^2\nu_1)} + \frac{\beta\nu_-\mu_+\mu_1^2}{(1-\beta\mu_1)(1-\beta^2\nu_1)}
\end{aligned}$$

and

$$
\begin{aligned}
&\mathbb{E}v_t^{\alpha_-}(\theta_0)v_t^{\beta}(\theta)\\
=&\sum_{i=1}^{\infty}\sum_{j=2}^{\infty}(j-1)\mathbb{E}\beta^{j-2}\frac{(\eta_{t-i}^-)^2}{a_0(\eta_{t-i})}\frac{1}{a_0(\eta_{t-1})\cdots a_0(\eta_{t-i+1})}\beta^{i-1}\frac{\alpha_+(\eta_{t-j}^+)^2+\alpha_-(\eta_{t-j}^-)^2}{a_0(\eta_{t-j})}\frac{1}{a_0(\eta_{t-1})\cdots a_0(\eta_{t-j+1})}\\
=&\sum_{j=2}^{\infty}(j-1)\beta^{2j-3}\mathbb{E}\frac{\alpha_+(\eta_{t-j}^-)^2(\eta_{t-j}^+)^2+\alpha_-(\eta_{t-j}^-)^4}{a_0(\eta_{t-j})^2}\frac{1}{a_0(\eta_{t-1})^2\cdots a_0(\eta_{t-j+1})^2}\\
&+\sum_{j=2}^{\infty}\sum_{i>j}(j-1)\beta^{i+j-3}\mathbb{E}\frac{(\eta_{t-i}^-)^2}{a_0(\eta_{t-i})}\frac{\alpha_+(\eta_{t-j}^+)^2+\alpha_-(\eta_{t-j}^-)^2}{a_0(\eta_{t-j})^2}\frac{1}{a_0(\eta_{t-1})^2\cdots a_0(\eta_{t-j+1})^2}\cdot\frac{1}{a_0(\eta_{t-j-1})\cdots a_0(\eta_{t-i+1})}\\
&+\sum_{j=2}^{\infty}\sum_{i=1}^{j-1}(j-1)\beta^{i+j-3}\mathbb{E}\frac{(\eta_{t-i}^-)^2}{a_0(\eta_{t-i})^2}\frac{\alpha_+(\eta_{t-j}^+)^2+\alpha_-(\eta_{t-j}^-)^2}{a_0(\eta_{t-j})}\frac{1}{a_0(\eta_{t-1})^2\cdots a_0(\eta_{t-i+1})^2}\cdot\frac{1}{a_0(\eta_{t-i-1})\cdots a_0(\eta_{t-j+1})}\\
=&\sum_{j=2}^{\infty}(j-1)\beta^{2j-3}(\alpha_+\delta_{+-}+\alpha_-\delta_-)\nu_1^{j-1}+\sum_{j=2}^{\infty}(j-1)\beta^{j-2}\sum_{i=j+1}^{\infty}\beta^{i-1}(\alpha_+\nu_++\alpha_-\nu_-)\mu_-\nu_1^{j-1}\mu_1^{i-j-1}\\
&+\sum_{j=2}^{\infty}(j-1)\beta^{j-2}\sum_{i=1}^{j-1}\beta^{i-1}\nu_-(\alpha_+\mu_++\alpha_-\mu_-)\nu_1^{i-1}\mu_1^{j-i-1}\\
=&\frac{\nu_1\beta(\alpha_+\delta_{+-}+\alpha_-\delta_-)}{(1-\beta^2\nu_1)^2}+\sum_{j=2}^{\infty}(j-1)\beta^{j-2}(\alpha_+\nu_++\alpha_-\nu_-)\mu_-\nu_1^{j-1}\frac{\beta^j}{1-\beta\mu_1}\\
&+\sum_{j=2}^{\infty}(j-1)\beta^{j-2}\mu_1^j(\alpha_+\mu_++\alpha_-\mu_-)\nu_-\frac{1-(\frac{\beta\nu_1}{\mu_1})^{j-1}}{1-\frac{\beta\nu_1}{\mu_1}}\\
=&\frac{\nu_1\beta(\alpha_+\delta_{+-}+\alpha_-\delta_-)}{(1-\beta^2\nu_1)^2}+\frac{(\alpha_+\nu_++\alpha_-\nu_-)\mu_-\beta^2\nu_1}{(1-\beta\mu_1)(1-\beta^2\nu_1)^2}+\frac{\mu_1^3\nu_-(\alpha_+\mu_++\alpha_-\mu_-)}{(\mu_1-\beta\nu_1)(1-\beta\mu_1)^2}-\frac{\mu_1^2\nu_-\beta\nu_1(\alpha_+\mu_++\alpha_-\mu_-)}{(1-\beta\mu_1)(1-\beta^2\nu_1)^2}.
\end{aligned}
$$

Third,

$$
\begin{aligned}
&\mathbb{E}v_t^{\alpha_+}(\theta_0)v_t^{\beta}(\theta)\\
=&\sum_{i=1}^{\infty}\sum_{j=2}^{\infty}(j-1)\mathbb{E}\beta^{j-2}\frac{(\eta_{t-i}^+)^2}{a_0(\eta_{t-i})}\frac{1}{a_0(\eta_{t-1})\cdots a_0(\eta_{t-i+1})}\beta^{i-1}\frac{\alpha_+(\eta_{t-j}^+)^2+\alpha_-(\eta_{t-j}^-)^2}{a_0(\eta_{t-j})}\frac{1}{a_0(\eta_{t-1})\cdots a_0(\eta_{t-j+1})}\\
=&\sum_{j=2}^{\infty}(j-1)\beta^{2j-3}\mathbb{E}\frac{\alpha_-(\eta_{t-j}^-)^2(\eta_{t-j}^+)^2+\alpha_+(\eta_{t-j}^+)^4}{a_0(\eta_{t-j})^2}\frac{1}{a_0(\eta_{t-1})^2\cdots a_0(\eta_{t-j+1})^2}\\
&+\sum_{j=2}^{\infty}\sum_{i>j}(j-1)\beta^{i+j-3}\mathbb{E}\frac{(\eta_{t-i}^+)^2}{a_0(\eta_{t-i})}\frac{\alpha_+(\eta_{t-j}^+)^2+\alpha_-(\eta_{t-j}^-)^2}{a_0(\eta_{t-j})^2}\frac{1}{a_0(\eta_{t-1})^2\cdots a_0(\eta_{t-j+1})^2}\cdot\frac{1}{a_0(\eta_{t-j-1})\cdots a_0(\eta_{t-i+1})}
\end{aligned}
$$

$$
\begin{aligned}
&+\sum_{j=2}^{\infty}\sum_{i=1}^{j-1}(j-1)\beta^{i+j-3}\mathbb{E}\frac{(\eta_{t-i}^{+})^2}{a_0(\eta_{t-i})^2}\frac{\alpha_+(\eta_{t-j}^{+})^2+\alpha_-(\eta_{t-j}^{-})^2}{a_0(\eta_{t-j})}\frac{1}{a_0(\eta_{t-1})^2\cdots a_0(\eta_{t-i+1})^2}\cdot\frac{1}{a_0(\eta_{t-i-1})\cdots a_0(\eta_{t-j+1})}\\
=&\sum_{j=2}^{\infty}(j-1)\beta^{2j-3}(\alpha_+\delta_++\alpha_-\delta_{+-})\nu_1^{j-1}+\sum_{j=2}^{\infty}(j-1)\beta^{j-2}\sum_{i=j+1}^{\infty}\beta^{i-1}(\alpha_+\nu_++\alpha_-\nu_-)\mu_+\nu_1^{j-1}\mu_1^{i-j-1}\\
&+\sum_{j=2}^{\infty}(j-1)\beta^{j-2}\sum_{i=1}^{j-1}\beta^{i-1}\nu_+(\alpha_+\mu_++\alpha_-\mu_-)\nu_1^{i-1}\mu_1^{j-i-1}\\
=&\frac{\nu_1\beta(\alpha_+\delta_++\alpha_-\delta_{+-})}{(1-\beta^2\nu_1)^2}+\frac{(\alpha_+\nu_++\alpha_-\nu_-)\mu_+\beta^2\nu_1}{(1-\beta\mu_1)(1-\beta^2\nu_1)^2}+\frac{\mu_1^3\nu_+(\alpha_+\mu_++\alpha_-\mu_-)}{(\mu_1-\beta\nu_1)(1-\beta\mu_1)^2}-\frac{\mu_1^2\nu_+\beta\nu_1(\alpha_+\mu_++\alpha_-\mu_-)}{(1-\beta\mu_1)(1-\beta^2\nu_1)^2}
\end{aligned}
$$

and

$$
\begin{aligned}
&\mathbb{E}(v_t^{\beta}(\theta))^2\\
=&\sum_{j=2}^{\infty}\sum_{i=2}^{\infty}\mathbb{E}(j-1)\beta^{j-2}(i-1)\beta^{i-2}\frac{\alpha_-(\eta_{t-j}^{-})^2+\alpha_+(\eta_{t-j}^{+})^2}{a_0(\eta_{t-j})}\frac{\alpha_-(\eta_{t-i}^{-})^2+\alpha_+(\eta_{t-i}^{+})^2}{a_0(\eta_{t-i})}\\
&\cdot\frac{1}{a_0(\eta_{t-1})\cdots a_0(\eta_{t-j+1})}\frac{1}{a_0(\eta_{t-1})\cdots a_0(\eta_{t-i+1})}\\
=&\sum_{j=2}^{\infty}(j-1)^2\beta^{2j-4}\mathbb{E}\frac{[\alpha_-(\eta_{t-j}^{-})^2+\alpha_+(\eta_{t-j}^{+})^2]^2}{a_0(\eta_{t-j})}\frac{1}{a_0(\eta_{t-1})^2\cdots a_0(\eta_{t-j+1})^2}+2\sum_{i>j}(j-1)(i-1)\beta^{j-2}\beta^{i-2}\\
&\cdot\frac{\alpha_-(\eta_{t-j}^{-})^2+\alpha_+(\eta_{t-j}^{+})^2}{a_0(\eta_{t-j})^2}\frac{\alpha_-(\eta_{t-i}^{-})^2+\alpha_+(\eta_{t-i}^{+})^2}{a_0(\eta_{t-i})}\frac{1}{a_0(\eta_{t-1})^2\cdots a_0(\eta_{t-j+1})^2}\frac{1}{a_0(\eta_{t-1})\cdots a_0(\eta_{t-i+1})}\\
=&(\alpha_-^2\delta_-+\alpha_+^2\delta_++2\alpha_+\alpha_-\delta_{+-})\sum_{j=2}^{\infty}(j-1)^2\nu_1^{j-1}\beta^{2j-4}\\
&+2\sum_{i>j}(j-1)(i-1)\beta^{i+j-4}(\alpha_-\nu_-+\alpha_+\nu_+)(\alpha_-\mu_-+\alpha_+\mu_+)\nu_1^{j-1}\mu_1^{i-j-1}\\
=&(\alpha_-^2\delta_-+\alpha_+^2\delta_++2\alpha_+\alpha_-\delta_{+-})\frac{\nu_1(1+\beta^2\nu_1)}{(1-\beta^2\nu_1)^3}\\
&+2(\alpha_-\nu_-+\alpha_+\nu_+)(\alpha_-\mu_-+\alpha_+\mu_+)\left[\frac{\beta^2\mu_1\nu_1}{(1-\beta^2\nu_1)^2(1-\mu_1\beta)^2}+\frac{2\beta^2\nu_1}{(1-\mu_1\beta)(1-\beta^2\nu_1)^3}\right].
\end{aligned}
$$

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